

Thus $\text{Der}(G, A) \approx \text{Hom}_{\mathbb{Z}G}(I, A)$. [Hint: A derivation $d: G \rightarrow A$ extends to an additive map $\bar{d}: \mathbb{Z}G \rightarrow A$ such that

$$\bar{d}(rs) = \bar{d}r \cdot \varepsilon(s) + r \cdot \bar{d}s$$

for all $r, s \in \mathbb{Z}G$, where $\varepsilon: \mathbb{Z}G \rightarrow \mathbb{Z}$ is the augmentation. The restriction of $\bar{d}$ to I is a module homomorphism such that $\bar{d}(g - 1) = dg$ for all $g \in G$.]

3. Let $F = F(S)$ be the free group generated by a set S .

(a) If A is an F -module and $(a_s)_{s \in S}$ is a family of elements of A , show that there is a unique derivation $d: F \rightarrow A$ such that $ds = a_s$ for all $s \in S$. [Hint: Derivations $F \rightarrow A$ correspond to splittings of $0 \rightarrow A \rightarrow A \rtimes F \rightarrow F \rightarrow 1$; now use the universal mapping property of F .]

(b) Deduce from (a) and exercise 2 that the augmentation ideal of $\mathbb{Z}F$ is a free $\mathbb{Z}F$ -module with basis $(Ds)_{s \in S}$. Thus we have reproved I.4.4 as well as exercise 3b of §II.5. As in the latter, we can now define the partial derivatives $\partial/\partial s: F \rightarrow \mathbb{Z}F$ by $Df = \sum_{s \in S} (\partial f/\partial s)Ds$ for $f \in F$.

(c) Reprove, from the present point of view, exercise 3c of §II.5. [Hint: The formula to be proved is true for the universal derivation, hence for any derivation.]

4. Let $G = F/R$, where $F = F(S)$ and R is the normal closure of a subset $T \subseteq F$.

(a) For any G -module A , show that derivations $G \rightarrow A$ correspond to derivations $d: F \rightarrow A$ such that $d(T) = 0$. [Hint: A splitting of $0 \rightarrow A \rightarrow A \rtimes G \rightarrow G \rightarrow 1$ can be defined by constructing a suitable map $F \rightarrow A \rtimes G$.]

(b) Using exercise 3, restate (a) as follows: Derivations $G \rightarrow A$ correspond to families $(a_s)_{s \in S}$ of elements of A such that $\sum_{s \in S} (\partial t/\partial s)a_s = 0$ for all $t \in T$, where $\partial t/\partial s$ is the image of $\partial t/\partial s$ under the quotient map $\mathbb{Z}F \rightarrow \mathbb{Z}G$.

(c) Using exercise 2, restate (b) as follows: Let I be the augmentation ideal of $\mathbb{Z}G$. Then there is an exact sequence

$$\mathbb{Z}G^{(T)} \xrightarrow{\partial_2} \mathbb{Z}G^{(S)} \xrightarrow{\partial_1} I \rightarrow 0,$$

where $\partial_1 e_s = \bar{s} - 1$ and $\partial_2 e_t = \sum_{s \in S} (\overline{\partial t/\partial s})e_s$. (This reproves the second part of exercise 3d of §II.5.)

(d) Make (c) more precise by proving that there is an exact sequence

$$(*) \quad 0 \rightarrow R_{ab} \xrightarrow{\theta} \mathbb{Z}G^{(S)} \xrightarrow{\partial} \mathbb{Z}G \xrightarrow{\varepsilon} \mathbb{Z} \rightarrow 0,$$

where $\partial e_s = \bar{s} - 1$ and $\theta(r \bmod [R, R]) = \sum_{s \in S} (\overline{\partial r/\partial s})e_s$. (This reproves II.5.4 and the first part of exercise 3d of §II.5.) [Hint: We have a free resolution

$$(**) \quad 0 \rightarrow I \rightarrow \mathbb{Z}F \xrightarrow{\varepsilon} \mathbb{Z} \rightarrow 0$$

of $\mathbb{Z}$ over $\mathbb{Z}F$, where I now denotes the augmentation ideal of $\mathbb{Z}F$. Taking R -coinvariants, we obtain a complex whose homology is $H_* R$. Since $I \approx \mathbb{Z}F^{(S)}$, this yields an exact sequence

$$0 \rightarrow H_1 R \xrightarrow{\varphi} \mathbb{Z}G^{(S)} \rightarrow \mathbb{Z}G \rightarrow \mathbb{Z} \rightarrow 0.$$

On the other hand, we can calculate H_*R from the standard (bar) resolution of $\mathbb{Z}$ over $\mathbb{Z}R$, and this is what we used to show $H_1R \approx R_{ab}$. Now it is easy to map the standard resolution to $(**)$, by $[\] \mapsto 1$ in dimension 0, $[r] \mapsto r - 1 = Dr$ in dimension 1, and $[r_1 | \cdots | r_n] \mapsto 0$ for $n > 1$. Deduce that there is a commutative diagram

$$\begin{array}{ccc} R & \xrightarrow{D|R} & I \\ \downarrow & & \downarrow \\ H_1R & \xrightarrow{\varphi} & \mathbb{Z}G^{(S)} \end{array}$$

where the vertical arrows are quotient maps. Since the composite $F \xrightarrow{D} I \rightarrow \mathbb{Z}G^{(S)}$ is a derivation such that $s \mapsto e_s$, the exactness of $(*)$ follows easily.]

3 The Classification of Extensions with Abelian Kernel

Let A be a fixed G -module. All extensions of G by A to be considered in this section will be assumed to give rise to the given action of G on A . To analyze an extension

$$(3.1) \quad 0 \rightarrow A \xrightarrow{i} E \xrightarrow{\pi} G \rightarrow 1,$$

we choose a set-theoretic cross-section of π , i.e., a function $s: G \rightarrow E$ such that $\pi s = \text{id}_G$. For simplicity, we will assume that s satisfies the *normalization condition*

$$(3.2) \quad s(1) = 1.$$

If s is a homomorphism, then the extension splits and we know its structure by 2.1. In the general case, however, there is a function $f: G \times G \rightarrow A$ which measures the failure of s to be a homomorphism. Indeed, for any $g, h \in G$, the elements $s(gh)$ and $s(g)s(h)$ of E both map to gh in G , so they differ by an element of $i(A)$. Thus we can define f by the equation

$$(3.3) \quad s(g)s(h) = i(f(g, h))s(gh).$$

Note that 3.2 implies that f is *normalized*, in the sense that

$$(3.4) \quad f(g, 1) = 0 = f(1, g)$$

for all $g \in G$. The function f is often called the *factor set* associated to 3.1 and s .

I claim now that the extension 3.1 can be completely recovered from the G -module structure on A and the function f . Indeed, since $s(G)$ is a set of coset representatives for $i(A)$ in E , we have a bijection $A \times G \rightarrow E$ given by $(a, g) \mapsto i(a)s(g)$. To compute the group law on $A \times G$ which makes this

bijection an isomorphism of groups, consider two elements $(a, g), (b, h)$ in $A \times G$. Using the commutation rule 1.2, we find

$$\begin{aligned} i(a)s(g)i(b)s(h) &= i(a)i(gb)s(g)s(h) \\ &= i(a + gb)i(f(g, h))s(gh) \\ &= i(a + gb + f(g, h))s(gh). \end{aligned}$$

Thus the group law on $A \times G$ is given by

$$(3.5) \quad (a, g)(b, h) = (a + gb + f(g, h), gh).$$

We will denote by E_f the set $A \times G$ with the product 3.5. Note that 3.5 looks like the product in $A \rtimes G$, "perturbed" by f .

Since $i(a) = i(a)s(1)$ for any $a \in A$, the composite $A \xrightarrow{i} E \approx E_f$ is the canonical inclusion

$$(3.6) \quad a \mapsto (a, 1).$$

And the composite $E_f \approx E \xrightarrow{\pi} G$ is obviously the canonical projection

$$(3.7) \quad (a, g) \mapsto g.$$

Thus the original extension 3.1 is equivalent to the extension

$$(3.8) \quad 0 \rightarrow A \rightarrow E_f \rightarrow G \rightarrow 1$$

defined by 3.5, 3.6, and 3.7 entirely in terms of G , A , and f .

It is natural to ask at this point whether we could start with an arbitrary function $f: G \times G \rightarrow A$ satisfying 3.4 and define a group E_f by means of 3.5. The answer is no. Indeed, if we define a product by 3.5 and compute the triple products $[(a, g)(b, h)](c, k)$ and $(a, g)[(b, h)(c, k)]$, we find that 3.5 is associative if and only if f satisfies the identity

$$(3.9) \quad f(g, h) + f(gh, k) = gf(h, k) + f(g, hk)$$

for all $g, h, k \in G$. If 3.9 holds, however, then we do in fact have a group. For (3.4) implies that $(0, 1)$ is a 2-sided identity, and it is easy to prove the existence of inverses. [Given $(a, g) \in E_f$, solve the equations $(a, g)(b, g^{-1}) = (0, 1)$ and $(b', g^{-1})(a, g) = (0, 1)$ for b and b' . One finds that (a, g) has left inverse $(-g^{-1}a - f(g^{-1}, g), g^{-1})$ and right inverse $(-g^{-1}a - g^{-1}f(g, g^{-1}), g^{-1})$. These two inverses are necessarily equal because of associativity.] Moreover, one checks easily that 3.6 and 3.7 define homomorphisms making the resulting sequence 3.8 exact, that 3.8 gives rise to the given action of G on A , and that the factor set associated to 3.8 (with the canonical cross-section $g \mapsto (0, g)$) is the original function f .

What we have established, then, is essentially a 1-1 correspondence

$$\left(\begin{array}{l} \text{extensions 3.1 with} \\ \text{a normalized section} \end{array} \right) \leftrightarrow \left(\begin{array}{l} \text{functions } G \times G \rightarrow A \\ \text{satisfying 3.4 and 3.9} \end{array} \right).$$

Now the identity 3.9 can be rewritten in a form which should look familiar:

$$(3.10) \quad gf(h, k) - f(gh, k) + f(g, hk) - f(g, h) = 0.$$

Indeed, f can be regarded as a 2-cochain of the standard complex $C^*(G, A)$ for computing $H^*(G, A)$ (cf. §III.1), and 3.10 says precisely that f is a cocycle. Moreover, 3.4 simply says that f is in the *normalized* cochain complex $C_N^*(G, A) \subseteq C^*(G, A)$. Thus

$$\left(\begin{array}{c} \text{extensions 3.1 with} \\ \text{a normalized section} \end{array} \right) \leftrightarrow \left(\begin{array}{c} \text{normalized 2-cocycles of} \\ G \text{ with coefficients in } A \end{array} \right)$$

Finally, I claim that changing the choice of the section in 3.1 corresponds precisely to modifying the cocycle f in $C_N^*(G, A)$ by a coboundary. In fact, an arbitrary normalized section of 3.1 is given by

$$(3.11) \quad g \mapsto i(c(g))s(g),$$

where $c: G \rightarrow A$ is a function such that $c(1) = 0$, i.e., c is an arbitrary element of $C_N^1(G, A)$. And we can easily compute the factor set associated to 3.11; for we have

$$\begin{aligned} i(c(g))s(g)i(c(h))s(h) &= i(c(g))i(gc(h))s(g)s(h) \\ &= i(c(g) + gc(h))i(f(g, h))s(gh) \\ &= i(c(g) + gc(h) + f(g, h) - c(gh))i(c(gh))s(gh). \end{aligned}$$

Thus the new factor set is

$$(g, h) \mapsto c(g) + gc(h) + f(g, h) - c(gh),$$

which is equal to $f + \delta c$, as claimed.

We have therefore proved:

(3.12) Theorem. *Let A be a G -module and let $\mathcal{E}(G, A)$ be the set of equivalence classes of extensions of G by A giving rise to the given action of G on A . Then there is a bijection*

$$\mathcal{E}(G, A) \approx H^2(G, A).$$

As a simple application of 3.12 and 2.3 we will prove the following theorem of group theory:

(3.13) Corollary. *Let E be a finite group of order mn , where m and n are relatively prime, and suppose that E contains an abelian normal subgroup A of order m . Then E contains subgroups of order n , and any two such subgroups are conjugate.*

PROOF. Let $G = E/A$ and consider the extension

$$0 \rightarrow A \rightarrow E \rightarrow G \rightarrow 1.$$

Since $|A|$ and $|G|$ are relatively prime, $H^2(G, A) = 0$ (cf. III.10.2). Thus $\mathcal{E}(G, A)$ contains a single element and hence the extension splits. This proves the existence part of the corollary. The uniqueness (up to conjugacy) follows similarly from 2.3, once one notes that any $\tilde{G} \subseteq E$ of order n must map isomorphically onto G and hence split the extension. $\square$

Remark. The corollary can be generalized to the case where A is non-abelian, cf. Zassenhaus [1958], IV.7. The existence part of this generalization, as well as the uniqueness if either A or E/A is assumed to be solvable, is proved by a straightforward inductive argument based on 3.13 and the Sylow theorems. One then completes the uniqueness proof by appealing to the Feit–Thompson theorem, which says that every group of odd order is solvable and hence that either A or E/A is solvable.

Further group-theoretic applications of 3.12 will be given in exercises 3–7 below and in §4.

EXERCISES

1. (Functorial properties of $\mathcal{E}(G, A)$)

(a) Given an extension $0 \rightarrow A \rightarrow E \rightarrow G \rightarrow 1$ and a group homomorphism $\alpha: G' \rightarrow G$, show that there is an extension $0 \rightarrow A \rightarrow E' \rightarrow G' \rightarrow 1$, characterized up to equivalence by the fact that it fits into a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & E & \longrightarrow & G & \longrightarrow & 1 \\ & & \parallel & & \uparrow & & \uparrow \alpha & & \\ 0 & \longrightarrow & A & \longrightarrow & E' & \longrightarrow & G' & \longrightarrow & 1. \end{array}$$

Deduce that α induces a map $\mathcal{E}(G, A) \rightarrow \mathcal{E}(G', A)$, which corresponds under the bijection of 3.12 to $H^2(\alpha, A): H^2(G, A) \rightarrow H^2(G', A)$. [Hint: Take E' to be the fiber-product (or “pull-back”) $E \times_G G'$, which by definition is the set of pairs $(e, g') \in E \times G'$ such that e and g' have the same image in G .]

(b) Given an extension $0 \rightarrow A \rightarrow E \rightarrow G \rightarrow 1$ and a homomorphism $f: A \rightarrow A'$ of G -modules, show that there is an extension $0 \rightarrow A' \rightarrow E' \rightarrow G \rightarrow 1$, characterized up to equivalence by the fact that it fits into a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & E & \longrightarrow & G & \longrightarrow & 1 \\ & & \downarrow f & & \downarrow & & \parallel & & \\ 0 & \longrightarrow & A' & \longrightarrow & E' & \longrightarrow & G & \longrightarrow & 1. \end{array}$$

Deduce that f induces a map $\mathcal{E}(G, A) \rightarrow \mathcal{E}(G, A')$, which corresponds under the bijection of 3.12 to $H^2(G, f): H^2(G, A) \rightarrow H^2(G, A')$. [Hint: Take E' to be the largest quotient of $A' \rtimes E$ such that the left-hand square above commutes.]

2. (Interpretation of $\delta: H^1 \rightarrow H^2$)

(a) Let $0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$ be a short exact sequence of G -modules. Given a derivation $d: G \rightarrow A''$, show that there is an extension $0 \rightarrow A' \rightarrow E \rightarrow G \rightarrow 1$, characterized by the fact that it fits into a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A' & \longrightarrow & A & \longrightarrow & A'' & \longrightarrow & 0 \\ & & \parallel & & \uparrow \bar{d} & & \uparrow d & & \\ 0 & \longrightarrow & A' & \longrightarrow & E & \longrightarrow & G & \longrightarrow & 1 \end{array}$$

with $\bar{d}$ a derivation. (The latter makes sense because A is an E -module via $E \rightarrow G$.) [Hint: Take E to be the set-theoretic pull-back, as in exercise 1a, regarded as a subgroup of $A \rtimes G$.]

(b) The construction in (a) gives a map $\text{Der}(G, A'') \rightarrow \mathcal{E}(G, A')$. Show that the diagram

$$\begin{array}{ccc} \text{Der}(G, A'') & \longrightarrow & \mathcal{E}(G, A') \\ \downarrow & & \downarrow \approx \\ H^1(G, A'') & \xrightarrow{\delta} & H^2(G, A') \end{array}$$

commutes. [Hint: A (set-theoretic) lifting of d to a function $G \rightarrow A$ yields a (set-theoretic) cross-section of $E \rightarrow G$. Now check definitions.]

3. Let G be a finite group. For any homomorphism $G \rightarrow \mathbb{Q}/\mathbb{Z}$ we can construct a central extension of G by $\mathbb{Z}$ by pulling back the canonical extension $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathbb{Q} & \longrightarrow & \mathbb{Q}/\mathbb{Z} & \longrightarrow & 0 \\ & & \parallel & & \uparrow & & \uparrow & & \\ 0 & \longrightarrow & \mathbb{Z} & \longrightarrow & E & \longrightarrow & G & \longrightarrow & 1. \end{array}$$

Thus we have a map $\varphi: \text{Hom}(G, \mathbb{Q}/\mathbb{Z}) \rightarrow \mathcal{E}(G, \mathbb{Z})$, where G acts trivially on $\mathbb{Z}$. Prove that φ is a bijection. [Hint: By exercise 2, φ can be identified with

$$\delta: H^1(G, \mathbb{Q}/\mathbb{Z}) \rightarrow H^2(G, \mathbb{Z}).]$$

4. (a) Let E be a group which contains a central subgroup C of finite index n . Prove that there is a homomorphism $E \rightarrow C$ whose restriction to C is the n -th power map. [Method 1: We have a central extension $1 \rightarrow C \rightarrow E \rightarrow G \rightarrow 1$ with $|G| = n$. Since n annihilates $H^2(G, C)$, the n -th power map $C \rightarrow C$ induces the zero-map on $H^2(G, C)$. Exercise 1b now implies that there is a diagram

$$\begin{array}{ccccccccc} 1 & \longrightarrow & C & \longrightarrow & E & \longrightarrow & G & \longrightarrow & 1 \\ & & \downarrow n & & \downarrow & & \parallel & & \\ 1 & \longrightarrow & C & \longrightarrow & C \times G & \longrightarrow & G & \longrightarrow & 1. \end{array}$$

Take the C -component of $E \rightarrow C \times G$. Method 2: Use the transfer map $H_1 E \rightarrow H_1 C = C$.]

(b) Deduce that the following two conditions on a finitely generated group E are equivalent:

- (i) The center C of E has finite index.
- (ii) The commutator subgroup $[E, E]$ of E is finite.

(Both of these conditions say, heuristically, that E is “almost abelian.”) [Hint: For (i) $\Rightarrow$ (ii), note that the map $E \rightarrow C$ of (a) has finite kernel. Conversely, (ii) implies that every generator of E has only finitely many conjugates, hence its centralizer has finite index.]

5. (a) Let E be a group which contains an infinite cyclic central subgroup of finite index. Prove that E is isomorphic to a semi-direct product $F \rtimes \mathbb{Z}$ with F finite. [Hint: Using either exercise 3 or exercise 4a, we can find a surjection $E \twoheadrightarrow \mathbb{Z}$ with finite kernel. This surjection must split since $\mathbb{Z}$ is free.]

(b) Deduce the following theorem of group theory: If E is a torsion-free group which has an infinite cyclic subgroup of finite index, then E is infinite cyclic. [Hint: Let $A \subseteq E$ be an infinite cyclic normal subgroup of finite index. Then E has a subgroup E' of index 1 or 2 such that A is central in E' . It follows from (a) that $E' \approx \mathbb{Z}$, so we have an extension $0 \rightarrow \mathbb{Z} \rightarrow E \rightarrow G \rightarrow 1$ with $|G| \leq 2$. If G acts non-trivially on $\mathbb{Z}$ then $H^2(G, \mathbb{Z}) = 0$ by direct computation. But then the extension splits, contradicting the assumption that E is torsion-free. Hence G acts trivially and $E \approx \mathbb{Z}$ by (a).]

- *6. Let E be a finitely generated torsion-free group which contains an abelian subgroup of finite index. Prove that E can be embedded as a discrete, co-compact subgroup of the group $\mathbb{R}^n \rtimes O_n$ of isometries of $\mathbb{R}^n$ for some n . [Hint: We have an extension $0 \rightarrow \mathbb{Z}^n \rightarrow E \rightarrow G \rightarrow 1$ with G finite. Now argue as in the hint to exercise 4a.]

7. (Universal central extensions). In this exercise all extensions will be assumed to be central and all coefficient modules for cohomology will be assumed to have trivial G -action. We will assume further that G is *perfect*, i.e., that $G = [G, G]$, or, equivalently, that $H_1 G = 0$. All (non-abelian) simple groups are perfect, for example.

(a) Show that there is a universal coefficient isomorphism

$$H^2(G, A) \approx \text{Hom}(H_2 G, A)$$

for any abelian group A .

(b) Deduce that there is a “universal” cohomology class $u \in H^2(G, H_2 G)$, with the following property: For any abelian group A and any $v \in H^2(G, A)$, there is a unique map $f: H_2 G \rightarrow A$ such that $v = H^2(G, f)u$. [Hint: Yoneda’s lemma (exercise 3a of §I.7) tells you how to describe the natural map $\text{Hom}(H_2 G, \text{---}) \xrightarrow{\sim} H^2(G, \text{---})$ of (a) in terms of its effect on $\text{id}_{H_2 G} \in \text{Hom}(H_2 G, H_2 G)$.]

(c) In view of 3.12 and exercise 1b, reinterpret (b) as saying that G admits a “universal central extension”

$$0 \rightarrow H_2 G \rightarrow E \rightarrow G \rightarrow 1,$$

characterized by the following property: Given any abelian group A and any central extension

$$0 \rightarrow A \rightarrow E' \rightarrow G \rightarrow 1,$$

there is a unique map $E \rightarrow E'$ making

$$\begin{array}{ccc} E & & \\ \downarrow & \searrow & \\ & G & \\ \uparrow & \swarrow & \\ E' & & \end{array}$$

commute. [Hint: To prove uniqueness, note that two such maps $E \rightarrow E'$ which induce the same map $H_2 G \rightarrow A$ must differ by a homomorphism $E \rightarrow A$ which factors through G . But G is perfect, so there are no non-trivial maps $G \rightarrow A$.]

Remark. In practice one often uses (c) to compute $H_2 G$. Namely, one produces a central extension of G with the universal property described in (c), and it then follows that the kernel of this extension is isomorphic to $H_2 G$. See Milnor [1971] for an interesting example of this ($G = SL_n(k)$) and further information about universal central extensions.

8. Let $0 \rightarrow A \xrightarrow{i} E \rightarrow G \rightarrow 1$ be a central extension with G abelian. The *commutator pairing* associated to the extension is the map $c: G \times G \rightarrow A$ defined by $c(g, h) = i^{-1}([\tilde{g}, \tilde{h}]) = i^{-1}(\tilde{g}\tilde{h}\tilde{g}^{-1}\tilde{h}^{-1})$, where $\tilde{g}$ and $\tilde{h}$ are lifts of g and h to E .

(a) Show that c is well-defined, $\mathbb{Z}$ -bilinear, and alternating. (The latter means that $c(g, g) = 0$; this implies, in view of bilinearity, that $c(g, h) = -c(h, g)$.) Thus c can be viewed as a map $\bigwedge^2 G \rightarrow A$, where $\bigwedge^2 G$, the second exterior power of G , is the quotient of $G \otimes G$ by the subgroup generated by the elements $g \otimes g$.

(b) If f is a factor set associated to the extension, show that $c(g, h) = f(g, h) - f(h, g)$.

(c) Let $\theta: H^2(G, A) \rightarrow \text{Hom}(\bigwedge^2 G, A)$ be the map which sends the class of a cocycle f to the alternating map $f(g, h) - f(h, g)$. Deduce from (b) that there is a bijection $\ker \theta \approx \mathcal{E}_{ab}(G, A)$, where $\mathcal{E}_{ab}(G, A)$ is the set of equivalence classes of abelian extensions of G by A .

See exercise 5 of §V.6 below for further results about θ .

4 Application: p -groups with a Cyclic Subgroup of Index p

As an illustration of the theory of group extensions, we will give in this section a classification of p -groups which contain a cyclic subgroup of index p , where p is a prime. (Such a subgroup is necessarily normal; cf. Hall [1959],

§4.2.) We will then use this classification to prove a theorem of Burnside (Theorem 4.3 below) which will be needed in §VI.9.

We begin by listing some examples of p -groups with a cyclic subgroup of index p :

(A) $\mathbb{Z}_q$ ($q = p^n$, $n \geq 1$).

(B) $\mathbb{Z}_q \times \mathbb{Z}_p$ ($q = p^n$, $n \geq 1$).

(C) $\mathbb{Z}_q \rtimes \mathbb{Z}_p$ ($q = p^n$, $n \geq 2$), where the canonical generator of $\mathbb{Z}_p$ acts on $\mathbb{Z}_q$ as multiplication by $1 + p^{n-1}$. (This makes sense because $(1 + p^{n-1})^p \equiv 1 \pmod{p^n}$ by the binomial theorem.)

If $p = 2$, there are three additional families:

(D) Dihedral 2-groups: Recall that for any integer $m \geq 2$, the dihedral group D_{2m} is defined to be $\mathbb{Z}_m \rtimes \mathbb{Z}_2$, where the generator of $\mathbb{Z}_2$ acts on $\mathbb{Z}_m$ as multiplication by -1 . If m is a power of 2, D_{2m} is a 2-group. Note that D_4 is of type (B) and D_8 is of type (C). For $m > 4$, however, D_{2m} is not isomorphic to any of the previous examples; this can be seen by computing abelianizations.

(E) Generalized quaternion 2-groups: Let $\mathbb{H}$ be the quaternion algebra $\mathbb{R} \oplus \mathbb{R}i \oplus \mathbb{R}j \oplus \mathbb{R}k$. For any integer $m \geq 2$, the generalized quaternion group Q_{4m} is defined to be the subgroup of the multiplicative group $\mathbb{H}^*$ generated by $x = e^{\pi i/m}$ and $y = j$. For example, Q_8 is simply the usual quaternion group $\{\pm 1, \pm i, \pm j, \pm k\}$. Note that x is of order $2m$ and that we have the relations $y^2 = x^m$ and $yx y^{-1} = x^{-1}$. It follows that the cyclic subgroup C generated by x is normal and of index 2 and hence that Q_{4m} has order $4m$. If m is a power of 2, Q_{4m} is a 2-group. For future reference we record some properties of Q_{4m} : (a) In the extension $0 \rightarrow C \rightarrow Q_{4m} \rightarrow \mathbb{Z}_2 \rightarrow 0$, the generator of $\mathbb{Z}_2$ acts as -1 on C . (b) Every element of $Q_{4m} - C$ is of order 4. (c) Q_8 has exactly three cyclic subgroups of index 2; Q_{4m} for $m > 2$ has a unique cyclic subgroup of index 2. [This follows from (b).] (d) Q_{4m} has a unique element of order 2. (e) The extension $0 \rightarrow C \rightarrow Q_{4m} \rightarrow \mathbb{Z}_2 \rightarrow 0$ does not split.

(F) $\mathbb{Z}_q \rtimes \mathbb{Z}_2$ ($q = 2^n$, $n \geq 3$), where the generator of $\mathbb{Z}_2$ acts on $\mathbb{Z}_q$ as multiplication by $-1 + 2^{n-1}$.

We now prove that this list is complete:

(4.1) Theorem. *If G is a p -group with a cyclic subgroup of index p , then G is isomorphic to one of the groups listed in (A)–(F) above.*

The proof will use the following lemma from elementary number theory:

(4.2) Lemma. *Let a be an integer such that $a^p \equiv 1 \pmod{p^n}$ for some $n \geq 2$. If p is odd then $a \equiv 1 \pmod{p^{n-1}}$. If $p = 2$ then $a \equiv \pm 1 \pmod{2^{n-1}}$.*

PROOF. Assuming, as we may, that $a \neq 1$, let $d(a)$ be the largest integer d such that $a \equiv 1 \pmod{p^d}$. Note that $d(a) \geq 1$ since $a \equiv a^p \equiv 1 \pmod{p}$, where the first congruence is by Fermat's little theorem. If p is odd or $d(a) \geq 2$,

then exercise 3a of §II.4 is applicable and gives $d(a^p) = d(a) + 1$. Thus $d(a) \geq n - 1$ in this case, as required. If $p = 2$ and $d(a) = 1$, then $a \equiv -1 \pmod{4}$. We can therefore apply the previous argument to $-a$ to conclude that $-a \equiv 1 \pmod{2^{n-1}}$. $\square$

PROOF OF THEOREM 4.1. By hypothesis we have an extension $0 \rightarrow \mathbb{Z}_q \rightarrow G \rightarrow H \rightarrow 1$, where $q = p^n$ for some $n \geq 0$ and $|H| = p$. If H acts trivially on $\mathbb{Z}_q$ then G is abelian, being generated by two commuting elements. It then follows from the theory of finite abelian groups that G is of type (A) or (B). Assume now that H acts non-trivially on $\mathbb{Z}_q$. We will use example 2 of §III.1 to compute $H^2(H, \mathbb{Z}_q)$. The action of H on $\mathbb{Z}_q$ is given by an embedding $H \hookrightarrow \mathbb{Z}_q^*$, where $\mathbb{Z}_q^*$ is the group of units of the ring $\mathbb{Z}_q = \mathbb{Z}/p^n\mathbb{Z}$, and clearly we must have $n \geq 2$. Suppose first that p is odd. Then Lemma 4.2 implies that the image of this embedding is $\{1 + b : b \in p^{n-1}\mathbb{Z}/p^n\mathbb{Z}\}$, which is a group of order p , generated by $1 + p^{n-1}$. In particular, H must have a generator which acts as $1 + p^{n-1}$, as in (C). Under this action of H on $\mathbb{Z}_q$, we have $\mathbb{Z}_q^H = \{x \in \mathbb{Z}_q : p^{n-1}x = 0\} = p\mathbb{Z}/p^n\mathbb{Z}$. On the other hand, the norm operator $\sum_{h \in H} h$ on $\mathbb{Z}_q$ is multiplication by $\sum (1 + b)$ ($b \in p^{n-1}\mathbb{Z}/p^n\mathbb{Z}$), which equals p . [$\sum b = 0$ because p is odd.] Thus the image of the norm operator is also equal to $p\mathbb{Z}/p^n\mathbb{Z}$. We conclude that $H^2(H, \mathbb{Z}_q) = 0$, so the extension splits and G is of type (C). Suppose now that $p = 2$. By Lemma 4.2 again, the image of $H \hookrightarrow \mathbb{Z}_q^*$ is a subgroup of $\{\pm 1 + b : b \in 2^{n-1}\mathbb{Z}/2^n\mathbb{Z}\}$. There are therefore three possibilities for the image $a \in \mathbb{Z}_q^*$ of the generator of H :

Case 1: $a = -1$. In this case $\mathbb{Z}_q^H = 2^{n-1}\mathbb{Z}/2^n\mathbb{Z}$ and the norm operator is zero, so $H^2(H, \mathbb{Z}_q) \approx \mathbb{Z}_2$. Thus there are precisely two inequivalent extensions of H by $\mathbb{Z}_q$ corresponding to this action of H on $\mathbb{Z}_q$. Since we have already produced two such extensions in (D) and (E), it follows that G is either of type (D) or type (E).

Case 2: $a = 1 + 2^{n-1}$. Then $\mathbb{Z}_q^H = 2\mathbb{Z}/2^n\mathbb{Z}$ and the norm operator is multiplication by $2 + 2^{n-1} = 2(1 + 2^{n-2})$. We may assume $n \geq 3$, since otherwise we are in case 1. Thus $1 + 2^{n-2} \in \mathbb{Z}_q^*$ and the image of the norm map is $2\mathbb{Z}/2^n\mathbb{Z}$. We therefore have $H^2(H, \mathbb{Z}_q) = 0$, so the extension splits and G is of type (C).

Case 3: $a = -1 + 2^{n-1}$, $n \geq 3$. Then $\mathbb{Z}_q^H = 2^{n-1}\mathbb{Z}/2^n\mathbb{Z}$ and the norm operator is multiplication by 2^{n-1} . Thus $H^2(H, \mathbb{Z}_q) = 0$, the extension splits, and G is of type (F). $\square$

Using this theorem, we can prove the following result (cf. Burnside [1911], §§104 and 105):

(4.3) Theorem. *If G is a p -group which has a unique subgroup of order p , then G is either a cyclic group or a generalized quaternion group.*

PROOF. Arguing by induction on $|G|$, we may assume that every proper subgroup of G is cyclic or generalized quaternion. Choose $H \triangleleft G$ of index p .

[Such an H always exists, cf. Hall [1959], §4.2.] If H is cyclic, then we are done by Theorem 4.1; for the only groups in the list (A)–(F) which have a unique subgroup of order p are the cyclic groups and the generalized quaternion groups. Suppose, then, that H is generalized quaternion (and hence that $p = 2$). I claim that H has a cyclic subgroup N of index 2 which is normal in G . Indeed, if $\mathcal{S}$ is the set of cyclic subgroups of H of index 2, then we know that $\text{card}(\mathcal{S})$ is odd (cf. statement (c) in the discussion of type (E) above); the conjugation action of the 2-group G on $\mathcal{S}$ must therefore fix some $N \in \mathcal{S}$, as claimed. Consider now the action of G/N on N . It is given by a homomorphism $G/N \rightarrow \mathbb{Z}_q^*$, where $q = |N| \geq 4$. Composing with the canonical projection $\mathbb{Z}_q^* \rightarrow \mathbb{Z}_4^* = \{\pm 1\}$, we obtain a surjection $G/N \rightarrow \{\pm 1\}$, whose kernel is a group K/N of order 2. Since the generator of K/N does not act as -1 on N , K cannot be generalized quaternion. Thus K is a cyclic subgroup of G of index 2, and we are done as before. $\square$

Remark. It is instructive to look at Burnside's proof of this theorem. Although neither cohomology theory nor the classification of group extensions had yet been developed, Burnside's proof contains, essentially, a direct proof that $\mathcal{E}(G, A) \approx A^G/NA$ for G finite cyclic, where N is the norm operator. [Explicitly, given an extension $0 \rightarrow A \rightarrow E \rightarrow G \rightarrow 1$, where G is cyclic of order n with generator t , Burnside lifts t to $\tilde{t} \in E$ and considers $a = \tilde{t}^n \in A$. Then a is in A^G and $a \bmod NA$ is the element of A^G/NA which classifies the extension.]

Finally, as one further illustration of the use of cohomology in group theory, we will determine all subgroups of a non-dihedral group of type (C):

(4.4) Proposition. *Let $G = \mathbb{Z}_q \rtimes \mathbb{Z}_p$ ($q = p^n, n \geq 2$) as in (C). If $p = 2$, assume $n \geq 3$. Let $A \subset \mathbb{Z}_q$ be the subgroup $p\mathbb{Z}/p^n\mathbb{Z}$ of index p .*

- (a) $A \rtimes \mathbb{Z}_p = A \times \mathbb{Z}_p$ is a non-cyclic abelian subgroup of G of index p .
- (b) Every element of G which is not in $A \times \mathbb{Z}_p$ generates a cyclic subgroup of index p . There are precisely p distinct such cyclic subgroups of index p .
- (c) Every proper subgroup H of G is either a subgroup of $A \times \mathbb{Z}_p$ or a cyclic subgroup of index p as in (b). In particular, H is abelian.

PROOF. (a) is obvious, since $\mathbb{Z}_p$ acts trivially on A . It is possible to prove (b) by a direct (and tedious) computation, but it is easier to proceed as follows. First, note that any homomorphism $G \rightarrow \mathbb{Z}_p$ must factor through the quotient $\mathbb{Z}_p \times \mathbb{Z}_p$ of G . It follows that the subgroups of G of index p correspond to the subgroups of $\mathbb{Z}_p \times \mathbb{Z}_p$ of index p , and there are precisely $p + 1$ of these. [They are the points of the projective line over the field with p elements.] Of these $p + 1$ subgroups of G of index p , one is $A \times \mathbb{Z}_p$ and I claim that the other p are all cyclic. Accepting this for the moment, we can easily prove (b) by a counting argument. For the p cyclic subgroups of

index p contain a total of $p(p^n - p^{n-1})$ generators, and none of these generators can be in $A \times \mathbb{Z}_p$. But $\text{card}(G - (A \times \mathbb{Z}_p)) = p^{n+1} - p^n = p(p^n - p^{n-1})$, so (b) follows at once. Since (c) is an immediate consequence of (b), it remains only to prove the claim.

In the proof of 4.1 we calculated the norm operator for the action of $\mathbb{Z}_p$ on $\mathbb{Z}_q$, and it follows from this computation that $H^1(\mathbb{Z}_p, \mathbb{Z}_q) = 0$. In view of Proposition 2.3, this means that the extension

$$0 \rightarrow \mathbb{Z}_q \rightarrow G \rightarrow \mathbb{Z}_p \rightarrow 0$$

has a unique splitting (up to conjugacy) and hence that G contains only two conjugacy classes of subgroups of order p . Suppose now that H is a non-cyclic subgroup of G of index p . Then we have an extension

$$0 \rightarrow A \rightarrow H \rightarrow \mathbb{Z}_p \rightarrow 0$$

with $\mathbb{Z}_p$ acting trivially on A (because $1 + p^{n-1} \equiv 1 \pmod{|A|}$). Thus H is abelian and non-cyclic, so H contains at least two subgroups of order p . Since H is normal in G , it follows that H contains both conjugacy classes of subgroups of G of order p . In particular, $0 \times \mathbb{Z}_p \subseteq H$ and hence $H = A \times \mathbb{Z}_p$, as claimed. $\square$

EXERCISES

1. If G is a generalized quaternion group, show that every subgroup of G is either cyclic or a generalized quaternion group. [This follows from Theorem 4.3, of course, but you can also check it directly.]
2. If G is a dihedral group, show that every subgroup of G is either cyclic or dihedral. [Hint: Every element of $G = \mathbb{Z}_q \rtimes \mathbb{Z}_2$ which is not in $\mathbb{Z}_q$ is of order 2.] If $|G| > 8$, show further that the non-cyclic abelian subgroups of G (i.e., the subgroups isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2 = D_4$) are not normal. D_8 , on the other hand, contains two non-cyclic abelian normal subgroups.
3. Let $G = \mathbb{Z}_q \rtimes \mathbb{Z}_2$ ($q = 2^n$, $n \geq 3$) as in (F). Let $A \subset \mathbb{Z}_q$ be the subgroup of order 2. Show that the only non-cyclic abelian subgroups of G are $A \times \mathbb{Z}_2$ and its conjugates. Show further that $A \times \mathbb{Z}_2$ is not normal in G . [Hint: If H is a non-cyclic proper subgroup of G , then we have an extension $0 \rightarrow H \cap \mathbb{Z}_q \rightarrow H \rightarrow \mathbb{Z}_2 \rightarrow 0$ with the generator of $\mathbb{Z}_2$ acting as -1 on $H \cap \mathbb{Z}_q$. This can only be abelian if $H \cap \mathbb{Z}_q = A$. Now calculate $H^1(\mathbb{Z}_2, \mathbb{Z}_q)$ and use this information as in the proof of 4.4.]
4. If G is a p -group such that every abelian normal subgroup is cyclic, show that G is of type (A), (D), (E), or (F), with $|G| \geq 16$ if G is of type (D). [Hint: Choose a maximal abelian normal subgroup of G , and consider the corresponding extension $0 \rightarrow \mathbb{Z}_q \rightarrow G \rightarrow H \rightarrow 1$, $q = p^n$. If $|H| \leq p$ then we are done by Theorem 4.1, since groups of type (B) and (C) have non-cyclic abelian normal subgroups. Suppose, then, that $|H| \geq p^2$, and consider the normal subgroups $H' \subseteq H$ of order p . Such an H' cannot act trivially on $\mathbb{Z}_q$, since the inverse image G' of H' in G would be an abelian normal

subgroup bigger than $\mathbb{Z}_q$. And H' cannot act as in (C) unless $p = 2$ and $n = 2$, since the non-cyclic subgroup of G' of index p (cf. 4.4) would be normal in G . So the only possibility is that $p = 2$ and that the non-trivial element of H' acts as -1 or $-1 + 2^{n-1}$ (with $n \geq 3$ in the latter case). But this is absurd, for the composite $H \rightarrow \mathbb{Z}_q^* \rightarrow \{\pm 1\}$ has a non-trivial kernel and we could take H' to be contained in this kernel.]

5 Crossed Modules and H^3 (Sketch)

We have seen in §§2 and 3 that H^1 and H^2 have concrete group-theoretic interpretations. It turns out that there are also group-theoretic interpretations of the functors H^n for $n \geq 3$, discovered independently by several people. (Many of the relevant references can be found in MacLane [1979].) We will confine ourselves here to a sketch of the theory in the case of H^3 . The essential ideas in this case go back to MacLane [1949], although he did not give the precise classification theorem 5.4 below.

Let E and N be groups. Suppose we are given an action of E on N , denoted $(e, n) \mapsto {}^e n$, as well as a homomorphism $\alpha: N \rightarrow E$ satisfying

$$(5.1) \quad \alpha({}^n n') = n n' n^{-1} \quad (n, n' \in N)$$

and

$$(5.2) \quad \alpha({}^e n) = e \alpha(n) e^{-1} \quad (e \in E, n \in N)$$

We then say that N (together with α and the action) is a *crossed module* over E .

The canonical example of a crossed module is that where E is the full automorphism group $\text{Aut } N$ and $\alpha(n)$ is the inner automorphism associated to n . Thus 5.1 is true by definition, and 5.2 is easily verified. Crossed modules also arise naturally in topology, and it is in this context that they were first introduced (Whitehead [1949]). Namely, if X is a topological space and Y is a subspace, then the relative homotopy group $\pi_2(X, Y)$ is a crossed module over $\pi_1 Y$, with α being the boundary map $\partial: \pi_2(X, Y) \rightarrow \pi_1 Y$. For another example, suppose E is the total space of a fibration with fiber F ; then one can make $\pi_1 F$ a crossed module over $\pi_1 E$.

Note that any ordinary (abelian) E -module can be viewed as a crossed module, with α being the trivial map. At the other extreme, any normal subgroup of E is a crossed module with E acting by conjugation and α being the inclusion.

Let N be a crossed module over E . We set $A = \ker \alpha$ and $G = \text{coker } \alpha$; the latter makes sense because $\text{im } \alpha$ is normal in E by 5.2. Note that A is central in N (by 5.1) and that the action of E on N induces an action of G on A . Thus we have a 4-term exact sequence

$$(5.3) \quad 0 \rightarrow A \xrightarrow{i} N \xrightarrow{\alpha} E \xrightarrow{\pi} G \rightarrow 1,$$

where A is a G -module. It turns out that exact sequences of this form are classified (up to a suitable equivalence relation) by $H^3(G, A)$.

More precisely, suppose we start with an arbitrary group G and an arbitrary G -module A . Consider all possible exact sequences of the form 5.3, where N is a crossed module over E such that the action of E on N induces the given action of G on A . We impose on these exact sequences the smallest equivalence relation such that

$$0 \rightarrow A \rightarrow N \rightarrow E \rightarrow G \rightarrow 1$$

is equivalent to

$$0 \rightarrow A \rightarrow N' \rightarrow E' \rightarrow G \rightarrow 1$$

whenever there is a commutative diagram

$$\begin{array}{ccccccc} & & & N & \longrightarrow & E & \\ & & \nearrow & \downarrow & & \downarrow & \searrow \\ 0 & \longrightarrow & A & & & & G \longrightarrow 1 \\ & & \searrow & \downarrow & & \downarrow & \nearrow \\ & & & N' & \longrightarrow & E' & \end{array}$$

in which the vertical arrows are compatible with the actions of E and E' on N and N' . [Warning: The vertical arrows need not be isomorphisms.] We then have:

(5.4) Theorem. *There is a 1-1 correspondence between equivalence classes as above and elements of $H^3(G, A)$.*

We omit the proof, which can be found in the references cited in MacLane [1979]. We will, however, explain how an exact sequence 5.3 gives rise to an element of $H^3(G, A)$. Choose a set-theoretic cross-section $s: G \rightarrow E$ of π . Its failure to be multiplicative is measured by a function $f: G \times G \rightarrow \ker \pi$ such that

$$(5.5) \quad s(g)s(h) = f(g, h)s(gh).$$

As in §3, the associativity of the product in E forces a “cocycle condition” on f , which takes the form

$$(5.6) \quad f(g, h)f(gh, k) = {}^{s(g)}f(h, k)f(g, hk),$$

where ${}^{s(g)}f(h, k) = s(g)f(h, k)s(g)^{-1}$. [5.6 is a non-abelian analogue of 3.9; it is proved by computing the triple product $s(g)s(h)s(k)$ in two different ways.]

Since $\ker \pi = \text{im } \alpha$, we can lift f to a function $F: G \times G \rightarrow N$, and we can ask whether F satisfies the analogue of 5.6 (which now involves the crossed-module action of E on N). The failure of F to do so is measured by a function $c: G \times G \times G \rightarrow A$ such that

$$(5.7) \quad {}^{s(g)}F(h, k)F(g, hk) = i(c(g, h, k))F(g, h)F(gh, k).$$

One then shows that c is a 3-cocycle, whose cohomology class is independent of the choices of s and F , and this cohomology class is the desired element of $H^3(G, A)$.

6 Extensions with Non-Abelian Kernel (Sketch)

The main references for this section are MacLane [1963] and Eilenberg–MacLane [1947].

Let N be a group and recall from §5 that N is an $\text{Aut}(N)$ -crossed module via the canonical map $\alpha: N \rightarrow \text{Aut}(N)$ and the canonical action of $\text{Aut}(N)$ on N . The kernel of α is the center C of N , and the cokernel of α is, by definition, the group $\text{Out}(N)$ of *outer automorphisms* of N . The resulting exact sequence

$$(6.1) \quad 0 \rightarrow C \rightarrow N \xrightarrow{\alpha} \text{Aut}(N) \rightarrow \text{Out}(N) \rightarrow 1$$

plays a fundamental role in the study of extensions

$$(6.2) \quad 1 \rightarrow N \xrightarrow{i} E \xrightarrow{\pi} G \rightarrow 1$$

with kernel N .

The first observation to make is that such an extension gives rise not to an action of G on N , but only to an “outer action,” i.e., a homomorphism $\psi: G \rightarrow \text{Out}(N)$. This is induced by the conjugation action of E on N :

$$\begin{array}{ccccc} N & \longrightarrow & E & \longrightarrow & G \\ \parallel & & \downarrow & & \downarrow \\ N & \longrightarrow & \text{Aut}(N) & \longrightarrow & \text{Out}(N). \end{array}$$

We therefore fix a homomorphism $\psi: G \rightarrow \text{Out}(N)$ and try to understand the set $\mathcal{E}(G, N, \psi)$ of equivalence classes of extensions giving rise to ψ . Note that it is not even obvious, for a given ψ , whether or not $\mathcal{E}(G, N, \psi)$ is non-empty. In particular, we do not have a semi-direct product $N \rtimes G$ in $\mathcal{E}(G, N, \psi)$ unless ψ lifts to a homomorphism $G \rightarrow \text{Aut}(N)$.

Suppose now that an extension 6.2 is given. As in §3, choose a set-theoretic cross-section $s: G \rightarrow E$ of π . This determines, as usual, a function $f: G \times G \rightarrow N$ measuring the failure of s to be a homomorphism. In addition, it determines a set-theoretic lifting $\varphi: G \rightarrow \text{Aut}(N)$ of ψ ; namely, $\varphi(g)$ is conjugation by $s(g)$. The functions f and φ are related by

$$(6.3) \quad \varphi(g)\varphi(h) = \alpha(f(g, h))\varphi(gh)$$

where α is as in 6.1. Moreover, f satisfies a “cocycle condition” (cf. 5.6)

$$(6.4) \quad f(g, h)f(gh, k) = {}^{\varphi(g)}f(h, k)f(g, hk).$$

Conversely, given f and φ satisfying 6.3 and 6.4, one can construct an extension

$$(6.5) \quad 1 \rightarrow N \rightarrow E_{f,\varphi} \rightarrow G \rightarrow 1,$$

where $E_{f,\varphi}$ is $N \times G$ with a product that can be written down explicitly in terms of f and φ . One sees in this way that extensions are classified by pairs (f, φ) as above, subject to some equivalence relation.

This description of $\mathcal{E}(G, N, \psi)$ can be drastically simplified once we observe that the lifting φ of ψ can be taken to be the same in all extensions under consideration; for by changing the choice of s , we can change our lifting φ to any other lifting. Thus we can fix φ and just consider the extensions $E_{f,\varphi}$, where f ranges over all “cocycles” relative to φ . It follows from 6.3 that any two such f ’s will differ by a function $G \times G \rightarrow C$, and the latter is an honest 2-cocycle of G with coefficients in C , where C is a G -module via ψ . One shows in this way that any two elements of $\mathcal{E}(G, N, \psi)$ have a well-defined “difference” in $H^2(G, C)$. This leads to:

(6.6) Theorem. *The set $\mathcal{E}(G, N, \psi)$ admits a free, transitive action by the abelian group $H^2(G, C)$. Hence either $\mathcal{E}(G, N, \psi) = \emptyset$ or else there is a bijection $\mathcal{E}(G, N, \psi) \approx H^2(G, C)$. This bijection depends on the choice of a particular element of $\mathcal{E}(G, N, \psi)$.*

To complete the classification, we must say when $\mathcal{E}(G, N, \psi) \neq \emptyset$. Recall from §5 that the sequence 6.1 yields an element $u \in H^3(\text{Out}(N), C)$. Applying the cohomology map $\psi^* = H^3(\psi, C)$, we obtain an element $\psi^*u \in H^3(G, C)$. I claim that ψ^*u is the “obstruction” to the existence of an element of $\mathcal{E}(G, N, \psi)$. To see this, we need the following explicit description of ψ^*u , which is easily deduced from §5: Choose a set-theoretic lifting $\varphi: G \rightarrow \text{Aut}(N)$ of ψ , and choose a function $f: G \times G \rightarrow N$ such that 6.3 holds. Then the failure of f to satisfy 6.4 is measured by a function $G \times G \times G \rightarrow C$ which is a 3-cocycle representing ψ^*u .

In particular, if $\mathcal{E}(G, N, \psi) \neq \emptyset$, then we have already seen that φ and f can be chosen to satisfy 6.4, so the 3-cocycle is zero. Conversely, if $\psi^*u = 0$, then we can change the choice of f so that 6.4 holds, and we can then construct an extension 6.5. This proves:

(6.7) Theorem. *A homomorphism $\psi: G \rightarrow \text{Out}(N)$ gives rise to an “obstruction” in $H^3(G, C)$, which vanishes if and only if $\mathcal{E}(G, N, \psi) \neq \emptyset$.*

This obstruction, incidentally, does not always vanish. Indeed, for any group G , any G -module C , and any element $v \in H^3(G, C)$, one can construct a group N with center C and a map $\psi: G \rightarrow \text{Out}(N)$ whose obstruction is equal to v .

Finally, we mention one special case where the theory of extensions is particularly simple:

(6.8) Corollary. *If N has a trivial center then there is exactly one extension of G by N (up to equivalence) corresponding to any homomorphism $G \rightarrow \text{Out}(N)$.*

This, of course, is an immediate consequence of 6.6 and 6.7, but it also admits an easy direct proof which uses no cohomology theory (cf. exercise 1 below).

EXERCISES

1. Give a direct proof of 6.8. [Hint: If N has trivial center then any extension of G by N fits into a diagram

$$\begin{array}{ccccccc}
 1 & \longrightarrow & N & \longrightarrow & E & \longrightarrow & G \longrightarrow 1 \\
 & & \parallel & & \downarrow & & \downarrow \\
 1 & \longrightarrow & N & \longrightarrow & \text{Aut}(N) & \longrightarrow & \text{Out}(N) \longrightarrow 1
 \end{array}$$

with exact rows. Such a diagram is necessarily a pull-back diagram (cf. §3, exercise 1a).]

2. Let $1 \rightarrow N \rightarrow E \rightarrow G \rightarrow 1$ be an extension of finite groups such that $|N|$ and $|G|$ are relatively prime. We proved in 3.13 that such an extension must split if N is abelian, and we remarked that this result could be generalized to the non-abelian case. One might hope to deduce this generalization directly from 6.6 in view of the vanishing of $H^2(G, C)$. Why doesn't this work?

CHAPTER V

Products

1 The Tensor Product of Resolutions

If G and G' are groups and M (resp. M') is a G -module (resp. G' -module), then $M \otimes M'$ is a $G \times G'$ -module in an obvious way: $(g, g') \cdot (m \otimes m') = gm \otimes g'm'$. Note that if M is projective over $\mathbb{Z}G$ and M' is projective over $\mathbb{Z}G'$ then $M \otimes M'$ is projective over $\mathbb{Z}[G \times G']$. In fact, it suffices to verify this in the case where $M = \mathbb{Z}G$ and $M' = \mathbb{Z}G'$, in which case the assertion follows from the obvious isomorphism $\mathbb{Z}G \otimes \mathbb{Z}G' \approx \mathbb{Z}[G \times G']$.

Now let $\varepsilon: F \rightarrow \mathbb{Z}$ (resp. $\varepsilon': F' \rightarrow \mathbb{Z}$) be a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$ (resp. $\mathbb{Z}G'$), and consider the complex $F \otimes F'$. This is a complex of projective $\mathbb{Z}[G \times G']$ -modules, and it is augmented over $\mathbb{Z}$ by $\varepsilon \otimes \varepsilon': F \otimes F' \rightarrow \mathbb{Z} \otimes \mathbb{Z} = \mathbb{Z}$. Moreover, $\varepsilon \otimes \varepsilon'$ is a weak equivalence. This follows, for instance, from I.7.6; for the latter shows that $\varepsilon: F \rightarrow \mathbb{Z}$ and $\varepsilon': F' \rightarrow \mathbb{Z}$ are homotopy equivalences if we ignore the G -action, so the same is true of $\varepsilon \otimes \varepsilon'$ (cf. exercise 7c of §I.0). [Alternatively, use the Künneth formula.]

We have therefore established:

(1.1) Proposition. *If $\varepsilon: F \rightarrow \mathbb{Z}$ and $\varepsilon': F' \rightarrow \mathbb{Z}$ are projective resolutions of $\mathbb{Z}$ over $\mathbb{Z}G$ and $\mathbb{Z}G'$, respectively, then $\varepsilon \otimes \varepsilon': F \otimes F' \rightarrow \mathbb{Z}$ is a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}[G \times G']$.*

This result should be thought of as the algebraic analogue of the obvious fact that the cartesian product of a $K(G, 1)$ and a $K(G', 1)$ is a $K(G \times G', 1)$.

(1.2) Corollary. *If $\varepsilon: F \rightarrow \mathbb{Z}$ and $\varepsilon': F' \rightarrow \mathbb{Z}$ are projective resolutions of $\mathbb{Z}$ over $\mathbb{Z}G$, then so is $\varepsilon \otimes \varepsilon': F \otimes F' \rightarrow \mathbb{Z}$, where G acts diagonally on $F \otimes F'$.*

This follows from 1.1 by restriction of scalars with respect to the “diagonal” embedding $G \rightarrow G \times G$, given by $g \mapsto (g, g)$.

Note that if $F = F'$ in 1.2 then we have two obvious maps $F \otimes F \rightarrow F$ of resolutions, namely, $F \otimes \varepsilon: F \otimes F \rightarrow F \otimes \mathbb{Z} = F$ and $\varepsilon \otimes F: F \otimes F \rightarrow \mathbb{Z} \otimes F = F$. There is no obvious map in the other direction, but we know from 1.7.5 that there exist augmentation-preserving G -chain maps $\Delta: F \rightarrow F \otimes F$, and that any two are homotopic. Any such map Δ will be called a *diagonal approximation*. In case F is the standard resolution, for example, there is a well-known diagonal approximation, called the Alexander–Whitney map. In terms of the homogeneous description of F , this map is given by

$$(1.3) \quad \Delta(g_0, \dots, g_n) = \sum_{p=0}^n (g_0, \dots, g_p) \otimes (g_p, \dots, g_n).$$

Translating this into the bar notation, we find

$$(1.4) \quad \Delta[g_1 | \dots | g_n] = \sum_{p=0}^n [g_1 | \dots | g_p] \otimes g_1 \cdots g_p [g_{p+1} | \dots | g_n].$$

EXERCISE

Let G be a finite cyclic group of order m with generator t and let F be the resolution 1.6.3. In particular, $F_n = \mathbb{Z}G$ for all $n \geq 0$. Let $\Delta: F \rightarrow F \otimes F$ be the map whose (p, q) -component $\Delta_{pq}: F_{p+q} \rightarrow F_p \otimes F_q$ is given by

$$\Delta_{pq}(1) = \begin{cases} 1 \otimes 1 & p \text{ even} \\ 1 \otimes t & p \text{ odd, } q \text{ even} \\ \sum_{0 \leq i < j \leq m-1} t^i \otimes t^j & p \text{ odd, } q \text{ odd.} \end{cases}$$

Verify that Δ is a diagonal approximation.

2 Cross-products

Let G, G', F , and F' , be as in the previous section. For any G -module M and G' -module M' there is an obvious map

$$(2.1) \quad (F \otimes_G M) \otimes (F' \otimes_{G'} M') \rightarrow (F \otimes F') \otimes_{G \times G'} (M \otimes M'),$$

given by $(x \otimes m) \otimes (x' \otimes m') \mapsto (x \otimes x') \otimes (m \otimes m')$. [One justifies this rigorously by noting that

$$(F \otimes_G M) \otimes (F' \otimes_{G'} M') = (F \otimes M)_G \otimes (F' \otimes M')_{G'},$$

which is a certain quotient of $F \otimes M \otimes F' \otimes M'$; making this quotient explicit, one can even see that the map 2.1 is an isomorphism.] If $z \in F \otimes_G M$

and $z' \in F' \otimes_{G'} M'$, then we will denote by $z \times z'$ the image of $z \otimes z'$ under this map. Clearly we have $\partial(z \times z') = \partial z \times z' + (-1)^p z \times \partial z'$ if $\deg z = p$; hence the product of two cycles is a cycle, whose homology class depends only on the classes of the given cycles. We therefore obtain, in view of 1.1, an induced product

$$H_p(G, M) \otimes H_q(G', M') \rightarrow H_{p+q}(G \times G', M \otimes M'),$$

called the *homology cross-product* and still denoted $z \otimes z' \mapsto z \times z'$. This product is well-defined, independent of the choice of the resolutions F and F' .

Similarly, given cochains $u \in \mathcal{H}om_G(F, M)$ and $u' \in \mathcal{H}om_{G'}(F', M')$, we define $u \times u' \in \mathcal{H}om_{G \times G'}(F \otimes F', M \otimes M')$ to be the tensor product of the maps u and u' as defined in exercise 7 of §I.0; thus $\langle u \times u', x \otimes x' \rangle = (-1)^{\deg u' \cdot \deg x} \langle u, x \rangle \otimes \langle u', x' \rangle$. It is easy to verify that $\delta(u \times u') = \delta u \times u' + (-1)^p u \times \delta u'$ if $\deg u = p$. [This is a special case of exercise 7a of I.0.] There is therefore an induced *cohomology cross-product*

$$H^p(G, M) \otimes H^q(G', M') \rightarrow H^{p+q}(G \times G', M \otimes M').$$

EXERCISES

1. Show that the map 2.1 is an isomorphism. Deduce, under suitable hypotheses, that there is a Künneth formula for computing $H_*(G \times G', M \otimes M')$ in terms of $H_*(G, M)$ and $H_*(G', M')$.
- *2. Give hypotheses under which the cochain cross-product

$$\mathcal{H}om_G(F, M) \otimes \mathcal{H}om_{G'}(F', M') \rightarrow \mathcal{H}om_{G \times G'}(F \otimes F', M \otimes M')$$

is an isomorphism, and deduce a cohomology Künneth formula.

3 Cup and Cap Products

The previous section dealt with “external” products, involving three groups G, G' , and $G \times G'$. In this section we will discuss *internal* products, involving the homology and/or cohomology of a single group G .

Given $u \in H^p(G, M)$ and $v \in H^q(G, N)$, we define the *cup product* of u and v (denoted $u \cup v$ or uv) to be the element $d^*(u \times v) \in H^{p+q}(G, M \otimes N)$, where $d: G \rightarrow G \times G$ is the diagonal map. Here $M \otimes N$ has the diagonal G -action (as it must if $d^*: H^*(G \times G, M \otimes N) \rightarrow H^*(G, M \otimes N)$ is to make sense).

Checking the definitions, we see that the cup product is induced by the following *cochain cup product*: Let F and F' be projective resolutions of $\mathbb{Z}$ over $\mathbb{Z}G$, and recall (1.2) that $F \otimes F'$ with diagonal G -action is also a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$. Given $u \in \mathcal{H}om_G(F, M)$ and $v \in \mathcal{H}om_G(F', N)$,

define $u \cup v$ to be $u \times v$ as defined in §2, but now regarded as an element of $\mathcal{H}om_G(F \otimes F', M \otimes N)$. Alternatively, if we prefer to use a single resolution F , then we choose a diagonal approximation $\Delta: F \rightarrow F \otimes F$ (cf. §1) and set $u \cup v = (u \times v) \circ \Delta \in \mathcal{H}om_G(F, M \otimes N)$ for $u \in \mathcal{H}om_G(F, M)$ and $v \in \mathcal{H}om_G(F, N)$. For example, if F is the bar resolution and Δ is the Alexander-Whitney map (1.4), then the product of $u \in C^p(G, M)$ and $v \in C^q(G, N)$ is the element $u \cup v \in C^{p+q}(G, M \otimes N)$ given by

$$(u \cup v)(g_1, \dots, g_{p+q}) = (-1)^{pq} u(g_1, \dots, g_p) \otimes g_1 \cdots g_p v(g_{p+1}, \dots, g_{p+q}).$$

We now list some formal properties of the cup product.

(3.1) *Dimension 0*: The cup product $H^0(G, M) \otimes H^0(G, N) \rightarrow H^0(G, M \otimes N)$ is the map $M^G \otimes N^G \rightarrow (M \otimes N)^G$ induced by the inclusions $M^G \hookrightarrow M$ and $N^G \hookrightarrow N$.

(3.2) *Naturality with respect to coefficient homomorphisms*: Given G -module maps $f: M \rightarrow M'$ and $g: N \rightarrow N'$ and elements $u \in H^*(G, M)$ and $v \in H^*(G, N)$, we have

$$(f \otimes g)_*(u \cup v) = f_* u \cup g_* v$$

in $H^*(G, M' \otimes N')$, where $f_* = H^*(G, f)$, etc.

3.1 and 3.2 are immediate from the definitions.

(3.3) *Compatibility with δ* : Let $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ be a short exact sequence of G -modules and let N be a G -module such that the sequence $0 \rightarrow M' \otimes N \rightarrow M \otimes N \rightarrow M'' \otimes N \rightarrow 0$ is exact. (For example, this holds for any N if $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ is split exact as a sequence of $\mathbb{Z}$ -modules.) Then we have $\delta(u \cup v) = \delta u \cup v$ for any $u \in H^p(G, M')$ and $v \in H^q(G, N)$. In other words, the square

$$\begin{array}{ccc} H^p(G, M') & \xrightarrow{\delta} & H^{p+1}(G, M') \\ \downarrow -\cup v & & \downarrow -\cup v \\ H^{p+q}(G, M' \otimes N) & \xrightarrow{\delta} & H^{p+q+1}(G, M' \otimes N) \end{array}$$

commutes.

PROOF. Consider the commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & C^*(G, M') & \longrightarrow & C^*(G, M) & \longrightarrow & C^*(G, M'') \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & C^*(G, M' \otimes N) & \longrightarrow & C^*(G, M \otimes N) & \longrightarrow & C^*(G, M'' \otimes N) \longrightarrow 0, \end{array}$$

where the vertical arrows are given by cup product on the right with a fixed cocycle in $C^q(G, N)$ representing v . These vertical maps commute with the coboundary operators δ in $C^*(G, -)$; for the formula $\delta(a \cup b) = \delta a \cup b + (-1)^{\deg a} a \cup \delta b$ reduces to $\delta(a \cup b) = \delta a \cup b$ if b is a cocycle. The result now follows from the naturality of connecting homomorphisms with respect to maps of short exact sequences (I.0.4). $\square$

Similarly, one proves:

(3.3') If $0 \rightarrow N' \rightarrow N \rightarrow N'' \rightarrow 0$ is a short exact sequence such that $0 \rightarrow M \otimes N' \rightarrow M \otimes N \rightarrow M \otimes N'' \rightarrow 0$ is exact, then $\delta(u \cup v) = (-1)^p u \cup \delta v$ in $H^{p+q+1}(G, M \otimes N')$ for any $u \in H^p(G, M)$, $v \in H^q(G, N')$.

3.3 and 3.3' allow one to use dimension-shifting arguments in the study of cup products. For we have seen (cf. exercise 3 of §III.7) that we can use a $\mathbb{Z}$ -split injection $M \rightarrow \bar{M}$ for the dimension-shifting argument.

(3.4) *Existence of identity*: The element $1 \in H^0(G, \mathbb{Z}) = \mathbb{Z}$ satisfies $1 \cup u = u = u \cup 1$ for all $u \in H^*(G, M)$, where we make the obvious identifications $\mathbb{Z} \otimes M = M = M \otimes \mathbb{Z}$ of coefficient modules.

This follows from the definitions together with the following two observations: (a) $1 \in H^0(G, \mathbb{Z})$ is represented by the augmentation map ε , regarded as a 0-cocycle in $\mathcal{H}om_G(F, \mathbb{Z})$; and (b) the maps $F \otimes \varepsilon$ and $\varepsilon \otimes F$ are maps of resolutions $F \otimes F \rightarrow F$ (cf. §1) and hence induce the "identity map" on cohomology. Alternatively, use the Alexander-Whitney formula.

(3.5) *Associativity*: Given $u_i \in H^*(G, M_i)$ ($i = 1, 2, 3$), we have $(u_1 u_2) u_3 = u_1 (u_2 u_3)$ in $H^*(G, M_1 \otimes M_2 \otimes M_3)$.

Indeed, associativity holds on the cochain level as an identity in $\mathcal{H}om_G(F \otimes F \otimes F, M_1 \otimes M_2 \otimes M_3)$. Alternatively, use the Alexander-Whitney formula.

(3.6) *Commutativity*: For any $u \in H^p(G, M)$, $v \in H^q(G, N)$, we have $uv = (-1)^{pq} t_*(vu)$, where $t: N \otimes M \rightarrow M \otimes N$ is the canonical isomorphism.

PROOF. Let $\tau: F \otimes F \rightarrow F \otimes F$ be the chain automorphism such that $\tau(x \otimes y) = (-1)^{\deg x \cdot \deg y} y \otimes x$, cf. exercise 5 of §I.0. We have a commutative diagram

$$\begin{array}{ccc} \mathcal{H}om_G(F, M) \otimes \mathcal{H}om_G(F, N) & \xrightarrow{\cup} & \mathcal{H}om_G(F \otimes F, M \otimes N) \\ \downarrow & & \downarrow \mathcal{H}om_G(\tau, t) \\ \mathcal{H}om_G(F, N) \otimes \mathcal{H}om_G(F, M) & \xrightarrow{\cup} & \mathcal{H}om_G(F \otimes F, N \otimes M) \end{array}$$

where the vertical arrow on the left is given by $u \otimes v \mapsto (-1)^{\deg u \cdot \deg v} v \otimes u$. [When you check the commutativity of this square, the signs may seem to

come out wrong at first; but don't forget that $\langle u, x \rangle = 0$ unless $\deg u = \deg x$.] Since τ is an augmentation-preserving chain map, it induces the identity in cohomology. Hence the vertical arrow on the right induces t_* in cohomology, and this yields 3.6. $\square$

It follows from 3.4–3.6 that $H^*(G, \mathbb{Z})$ is an anti-commutative graded ring and that $H^*(G, M)$ is a graded module over this ring. More generally, if k is an arbitrary commutative ring (with trivial G -action, for simplicity), then we have a product

$$H^*(G, k) \otimes H^*(G, k) \xrightarrow{\smile} H^*(G, k \otimes k) \rightarrow H^*(G, k)$$

which makes $H^*(G, k)$ a graded anti-commutative k -algebra. (The unlabelled arrow above is induced by the multiplication map $k \otimes k \rightarrow k$.) Similarly, $H^*(G, M)$ is an $H^*(G, k)$ -module if M is a kG -module.

(3.7) *Naturality with respect to group homomorphisms:* Given $\alpha: H \rightarrow G$, we have $\alpha^*(u \cup v) = \alpha^*u \cup \alpha^*v$ for any $u \in H^*(G, M)$, $v \in H^*(G, N)$.

This is immediate from the definitions.

As a special case of 3.7, $\alpha^*: H^*(G, k) \rightarrow H^*(H, k)$ is a ring homomorphism.

(3.8) *Transfer formula:* Suppose $H \subseteq G$ is a subgroup of finite index. For any $u \in H^*(G, M)$ and $v \in H^*(H, N)$, we have

$$\text{cor}_H^G(\text{res}_H^G(u) \cup v) = u \cup \text{cor}_H^G v.$$

This says, in particular, that the transfer map $H^*(H, k) \rightarrow H^*(G, k)$ is a homomorphism of $H^*(G, k)$ -modules, where $H^*(H, k)$ is regarded as an $H^*(G, k)$ -module via the restriction homomorphism $H^*(G, k) \rightarrow H^*(H, k)$.

PROOF OF 3.8. Let F be a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$; we will prove the stated formula on the cochain level. Given $u \in \mathcal{H}om(F, M)^G$ and $v \in \mathcal{H}om(F, N)^H$, we have, in $\mathcal{H}om(F \otimes F, M \otimes N)^G$:

$$\begin{aligned} \text{cor}_H^G(\text{res}_H^G(u) \cup v) &= \sum_{g \in G/H} g \cdot (u \otimes v) \\ &= \sum_{g \in G/H} u \otimes gv \quad [\text{because } u \text{ is } G\text{-invariant}] \\ &= u \otimes \sum_{g \in G/H} gv \\ &= u \cup \text{cor}_H^G v. \end{aligned} \quad \square$$

There is a second internal product, called the *cap product*, which is useful in the study of duality (cf. §§VI.7 and VIII.10). If F is a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$, then there is a chain map

$$\gamma: \mathcal{H}om_G(F, M) \otimes ((F \otimes F) \otimes_G N) \rightarrow F \otimes_G (M \otimes N)$$

given by $u \otimes (x \otimes y \otimes n) \mapsto (-1)^{\deg u \cdot \deg x} x \otimes u(y) \otimes n$. [The reader can check directly that γ is well-defined and is a chain map. Alternatively, we can appeal to the exercises of §I.0: According to exercise 7b of the latter, we have a chain map $\mathcal{H}om_G(F, M) \rightarrow \mathcal{H}om((F \otimes N) \otimes_G F, (F \otimes N) \otimes_G M)$ given by $u \mapsto \text{id}_{F \otimes N} \otimes u$. This corresponds by exercise 6b of §I.0 to a chain map

$$\mathcal{H}om_G(F, M) \otimes ((F \otimes N) \otimes_G F) \rightarrow (F \otimes N) \otimes_G M,$$

which is precisely γ , modulo the identifications $(F \otimes N) \otimes_G F = (F \otimes F) \otimes_G N$ and $(F \otimes N) \otimes_G M = F \otimes_G (M \otimes N)$. These identifications can be derived by manipulating triple tensor products as in the proof of III.2.2.]

For $u \in \mathcal{H}om_G(F, M)^p = \mathcal{H}om_G(F, M)_{-p}$ and $z \in (F \otimes F)_q \otimes_G N$, $\gamma(u \otimes z)$ is an element of $F_{q-p} \otimes_G (M \otimes N)$, denoted $u \cap z$ and called the *cap product* of u and z . The same notation and terminology are used for the induced product

$$H^p(G, M) \otimes H_q(G, N) \rightarrow H_{q-p}(G, M \otimes N).$$

As with the cup product, one can use a diagonal approximation $\Delta: F \rightarrow F \otimes F$ to compute the cap product in terms of a single resolution F . Namely, one composes γ with the map

$$\text{id} \otimes (\Delta \otimes \text{id}): \mathcal{H}om_G(F, M) \otimes (F \otimes_G N) \rightarrow \mathcal{H}om_G(F, M) \otimes ((F \otimes F) \otimes_G N).$$

The cap product, which may seem strange at first, can be motivated by the fact that it is *adjoint* to the cup product, in a sense which we now explain. Consider the “evaluation map”

$$\mathcal{H}om_G(F, M) \otimes (F \otimes_G N) \rightarrow M \otimes_G N,$$

given by $u \otimes (x \otimes n) \mapsto u(x) \otimes n$. We denote by $\langle u, z \rangle$ the image of $u \otimes z$ under this map. [Except for the fact that we are carrying along the factor N , this is the same as the evaluation map introduced in exercise 3 of §I.0.] The evaluation map is a chain map, i.e., $\langle \delta u, z \rangle + (-1)^{\deg u} \langle u, \partial z \rangle = 0$; so there is an induced pairing

$$H^p(G, M) \otimes H_q(G, N) \rightarrow M \otimes_G N,$$

still denoted $\langle \cdot, \cdot \rangle$, which is independent of the choice of resolution. One now checks, directly from the definitions, that the following adjunction formula holds for any $u \in H^p(G, M_1)$, $v \in H^q(G, M_2)$, $z \in H_{p+q}(G, M_3)$:

$$(3.9) \quad \langle u \cup v, z \rangle = \langle u, v \cap z \rangle.$$

(Note: Both sides of this equation are in $(M_1 \otimes M_2 \otimes M_3)_G$.) In particular, taking $u = 1 \in H^0(G, \mathbb{Z})$, we find:

(3.10) For any $v \in H^q(G, M)$ and $z \in H_q(G, N)$, we have $v \cap z = \langle v, z \rangle$ in $H_0(G, M \otimes N) = M \otimes_G N$.

The cap product has properties analogous to 3.1–3.8. We leave it to the interested reader to write these down.

EXERCISES

- Given $m \in H^0(G, M) = M^G$ and $u \in H^q(G, N)$, show that $m \cup u = f_m(u)$, where $f_m: H^*(G, N) \rightarrow H^*(G, M \otimes N)$ is induced by the coefficient homomorphism $n \mapsto m \otimes n$. [Hint: Use 3.2 and 3.4.] State and prove a similar interpretation of the cap product $H^0(G, M) \otimes H_q(G, N) \rightarrow H_q(G, M \otimes N)$.
- Using the diagonal approximation given in the exercise of §1, compute all cup products in $H^*(G, \mathbb{Z})$ if G is finite cyclic.
- Let G be a finite group which acts freely on S^{2k-1} as in §1.6.
 - For any G -module M , show that there is an iterated coboundary map $d: H^i(G, M) \rightarrow H^{i+2k}(G, M)$ which is an isomorphism for $i > 0$ and an epimorphism for $i = 0$. [Hint: Tensor the sequence I.6.1 with M , break it up into short exact sequences, and use the dimension-shifting argument.]
 - Show that there is an element $u \in H^{2k}(G, \mathbb{Z})$ such that the “periodicity map” d of (a) is given by $d(v) = u \cup v$ for all $v \in H^*(G, M)$. [Hint: By 3.3, $d(w \cup v) = d(w) \cup v$ for any $w \in H^*(G, \mathbb{Z})$, $v \in H^*(G, M)$; now set $w = 1$.]
 - Using (b), calculate the ring structure on $H^*(G, \mathbb{Z})$ for G finite cyclic.
- Let G be cyclic of order n . For any $m \in \mathbb{Z}_n$ there is an endomorphism $\alpha(m)$ of G , given by $\alpha(m)g = g^m$. Calculate $\alpha(m)^*: H^*(G, \mathbb{Z}) \rightarrow H^*(G, \mathbb{Z})$. [Hint: In view of what you know about the ring structure on $H^*(G, \mathbb{Z})$ from exercise 2 and/or 3c, it suffices to calculate $\alpha(m)^*$ on $H^2(G, \mathbb{Z})$. This can be done non-computationally by using the universal coefficient isomorphism $H^2(G, \mathbb{Z}) \approx \text{Ext}(H_1 G, \mathbb{Z})$ or by using the interpretation of H^2 in terms of group extensions.]
- Using exercise 4 and Theorem III.10.3, calculate the integral cohomology of the symmetric group on 3 letters.

4 Composition Products

We observed several times in Chapter III that there are a number of chain complexes that can be used to compute $H_*(G, \mathbb{Z})$ and $H^*(G, \mathbb{Z})$. Here is one more example of this:

(4.1) Lemma. Let $\varepsilon: F \rightarrow \mathbb{Z}$ and $\varepsilon': F' \rightarrow \mathbb{Z}$ be resolutions of $\mathbb{Z}$ over $\mathbb{Z}G$ such that F is projective and F' is $\mathbb{Z}$ -free. For any G -module M , the map $\varepsilon' \otimes M: F' \otimes M \rightarrow \mathbb{Z} \otimes M = M$ induces weak equivalences $F \otimes_G (F' \otimes M) \rightarrow F \otimes_G M$ and $\mathcal{H}om_G(F, F' \otimes M) \rightarrow \mathcal{H}om_G(F, M)$. Hence

$$H_*(G, M) \approx H_*(F \otimes_G (F' \otimes M))$$

and

$$H^*(G, M) \approx H^*(\mathcal{H}om_G(F, F' \otimes M)).$$

PROOF. We already noted in §III.2 that $\varepsilon' \otimes M: F' \otimes M \rightarrow M$ is a weak equivalence. The lemma therefore follows from I.8.5 and I.8.6. $\square$

Note that the last isomorphism in 4.1 can be written in the form $H^*(G, M) \approx [F, F' \otimes M]^*$, where $[F, F' \otimes M]^n = [F, F' \otimes M]_{-n}$ is the group of homotopy classes of chain maps of degree $-n$ from F' to $F' \otimes M$ (cf. §I.0). The significance of this is that it allows us to construct products by composition of chain maps. Taking $M = \mathbb{Z}$ and $F = F'$, for instance, the isomorphism takes the form $H^*(G, \mathbb{Z}) \approx [F, F]^*$. Since homotopy classes of chain maps can be composed, we obtain a *composition product*

$$H^*(G, \mathbb{Z}) \otimes H^*(G, \mathbb{Z}) \rightarrow H^*(G, \mathbb{Z}).$$

More generally, let $\varepsilon: F \rightarrow \mathbb{Z}$, $\varepsilon': F' \rightarrow \mathbb{Z}$, and $\varepsilon'': F'' \rightarrow \mathbb{Z}$ be resolutions of $\mathbb{Z}$ over $\mathbb{Z}G$ such that F and F' are projective and F'' is $\mathbb{Z}$ -free. [In practice, we will either take F'' projective or $F'' = \mathbb{Z}$.] For any G -modules M, N , there is a cochain product

$$(4.2) \quad \mathcal{H}om_G(F', F'' \otimes M) \otimes \mathcal{H}om_G(F, F' \otimes N) \rightarrow \mathcal{H}om_G(F, F'' \otimes M \otimes N)$$

given by $u \otimes v \mapsto (u \otimes \text{id}_N) \circ v$ for $u \in \mathcal{H}om_G(F', F'' \otimes M)$, $v \in \mathcal{H}om_G(F, F' \otimes N)$:

$$F \xrightarrow{u} F' \otimes N \xrightarrow{u \otimes \text{id}} F'' \otimes M \otimes N.$$

One easily verifies that 4.2 is a chain map (cf. exercise 4 of §I.0); in view of 4.1, there is an induced cohomology product

$$(4.3) \quad H^*(G, M) \otimes H^*(G, N) \rightarrow H^*(G, M \otimes N),$$

called the *composition product*. It is well-defined, independent of the choices of resolution.

Similarly, there is a product

$$(4.4) \quad \mathcal{H}om_G(F', F'' \otimes M) \otimes (F \otimes_G (F' \otimes N)) \rightarrow F \otimes_G (F'' \otimes M \otimes N)$$

given by

$$u \otimes x \otimes x' \otimes n \mapsto (-1)^{\deg u \cdot \deg x} x \otimes u(x') \otimes n$$

for $u \in \mathcal{H}om_G(F', F'' \otimes M)$, $x \in F$, $x' \in F'$, $n \in N$. One verifies as in the definition of the cap product that 4.4 is chain map; in view of 4.1, there is an induced product

$$(4.5) \quad H^*(G, M) \otimes H_*(G, N) \rightarrow H_*(G, M \otimes N).$$

The following result, although not unexpected, is useful:

(4.6) Theorem. *The products 4.3 and 4.5 coincide with the cup and cap products.*

PROOF. Consider the cochain composition product 4.2 with $F = F'$ and $F'' = \mathbb{Z}$:

$$(*) \quad \mathcal{H}om_G(F, M) \otimes \mathcal{H}om_G(F, F \otimes N) \rightarrow \mathcal{H}om_G(F, M \otimes N).$$

This induces the product 4.3 via the weak equivalence $\alpha = \mathcal{H}om_G(F, \varepsilon \otimes N): \mathcal{H}om_G(F, F \otimes N) \rightarrow \mathcal{H}om_G(F, N)$ of 4.1. In order to compare 4.3 to the cup product, we would like to find a weak inverse of α , i.e., a map $\alpha': \mathcal{H}om_G(F, N) \rightarrow \mathcal{H}om_G(F, F \otimes N)$ which induces in cohomology the isomorphism inverse to that induced by α . This would allow us to convert $(*)$ to a cochain product

$$(**) \quad \mathcal{H}om_G(F, M) \otimes \mathcal{H}om_G(F, N) \rightarrow \mathcal{H}om_G(F, M \otimes N)$$

by composing $(*)$ with

$$\text{id} \otimes \alpha': \mathcal{H}om_G(F, M) \otimes \mathcal{H}om_G(F, N) \rightarrow \mathcal{H}om_G(F, M) \otimes \mathcal{H}om_G(F, F \otimes N).$$

It turns out that a diagonal approximation $\Delta: F \rightarrow F \otimes F$ gives rise to such an α' . Namely, we take α' to be the composite

$$\mathcal{H}om_G(F, N) \xrightarrow{\text{id}_F \otimes -} \mathcal{H}om_G(F \otimes F, F \otimes N) \xrightarrow{- \circ \Delta} \mathcal{H}om_G(F, F \otimes N),$$

i.e., $\alpha'(u) = (\text{id}_F \otimes u) \circ \Delta$. Computing $\alpha\alpha': \mathcal{H}om_G(F, N) \rightarrow \mathcal{H}om_G(F, N)$, one finds that it is the map

$$u \mapsto (\varepsilon \otimes \text{id}_N) \circ (\text{id}_F \otimes u) \circ \Delta = (\varepsilon \otimes u) \circ \Delta.$$

But this is simply $u \mapsto \varepsilon \cup u$; since the cocycle ε represents $1 \in H^0(G, \mathbb{Z})$, it follows that α' is indeed a weak inverse of α .

We now have a cochain product $(**)$ which induces the composition product in cohomology, and it is given explicitly by

$$\begin{aligned} u \otimes v &\mapsto (u \otimes \text{id}_N) \circ \alpha'(v) = (u \otimes \text{id}_N)(\text{id}_F \otimes v)\Delta \\ &= (u \otimes v)\Delta \\ &= u \cup v, \end{aligned}$$

where the latter is the cochain cup product defined via Δ . Thus the composition product in cohomology equals the cup product.

The proof that 4.5 equals the cap product is similar, and even easier. This time we consider the map 4.4, with $F = F'$ and $F'' = \mathbb{Z}$, and we seek a weak inverse $\beta': F \otimes_G N \rightarrow F \otimes_G (F \otimes N)$ of the weak equivalence $\beta = F \otimes \varepsilon \otimes N: F \otimes_G (F \otimes N) \rightarrow F \otimes_G N$ of 4.1. I claim we can take $\beta' = \Delta \otimes N$. For the composite $(F \otimes \varepsilon \otimes N) \circ (\Delta \otimes N): F \otimes_G N \rightarrow F \otimes_G N$ is a map of the form $\tau \otimes N$, where $\tau: F \rightarrow F$ is an augmentation-preserving chain map, hence it is homotopic to the identity. We now use β' to convert 4.4 to a product

$$\mathcal{H}om_G(F, M) \otimes (F \otimes_G N) \rightarrow F \otimes_G (M \otimes N),$$

and it is transparent that this map coincides with the chain level cap product defined by Δ . $\square$

We close by remarking that the methods of this section can be used to define composition products

$$(4.7) \quad \text{Ext}_G^*(M', M'') \otimes \text{Ext}_G^*(M, M') \rightarrow \text{Ext}_G^*(M, M'')$$

and

$$(4.8) \quad \text{Ext}_G^*(M', M'') \otimes \text{Tor}_*^G(M, M') \rightarrow \text{Tor}_*^G(M, M'')$$

for any G -modules M, M', M'' . One uses suitable resolutions F, F', F'' of the three modules and one defines maps $\mathcal{H}om_G(F', F'') \otimes \mathcal{H}om_G(F, F') \rightarrow \mathcal{H}om_G(F, F'')$ and $\mathcal{H}om_G(F', F'') \otimes (F \otimes_G F') \rightarrow F \otimes_G F''$ similar to 4.2 and 4.4. The products 4.7 and 4.8 are obviously closely related to 4.3 and 4.5, but it does not seem that either type is a special case of the other. For instance 4.3, which we now know is the same as the cup product, can be written in the form

$$\text{Ext}_G^*(\mathbb{Z}, M) \otimes \text{Ext}_G^*(\mathbb{Z}, N) \rightarrow \text{Ext}_G^*(\mathbb{Z}, M \otimes N).$$

This fits into the framework of 4.7 in the important special case where $N = \mathbb{Z}$, but not in general. In the other direction, we know that the Ext groups in 4.7 can be expressed as cohomology groups if M and M' are $\mathbb{Z}$ -free (cf. III.2.2), so we expect 4.7 to be describable in terms of cup products in this case, but not in general.

Note, incidentally, that 4.7 and 4.8 can be defined for Ext and Tor over arbitrary rings, not just group rings. One need only be careful about which modules are left modules and which are right modules.

EXERCISES

1. Carry out the details of the definitions of 4.7 and 4.8.
2. Describe 4.7 in terms of cup products if M and M' are $\mathbb{Z}$ -free. Describe 4.8 in terms of cap products if M' is $\mathbb{Z}$ -free and M is $\mathbb{Z}$ -torsion-free. [Hint: Use Theorem 4.6.]

5 The Pontryagin Product

If G is an *abelian* group then the multiplication map $G \times G \rightarrow G$ is a group homomorphism. Using this map, we will define an internal homology product, called the Pontryagin product. For simplicity, we will take the coefficient module to be a commutative ring k , with trivial G -action. See exercise 1 below for a more general product.

The *Pontryagin product* on $H_*(G, k)$ for G abelian is defined to be the composite

$$H_*(G, k) \otimes H_*(G, k) \xrightarrow{\times} H_*(G \times G, k \otimes k) \xrightarrow{\mu_*} H_*(G, k),$$

where $\mu: (G \times G, k \otimes k) \rightarrow (G, k)$ is the multiplication map $((g, g') \mapsto gg', \lambda \otimes \lambda' \mapsto \lambda\lambda')$. To compute this product in terms of a projective resolution F of $\mathbb{Z}$ over $\mathbb{Z}G$, we need to choose an augmentation-preserving chain map $\tau: F \otimes F \rightarrow F$ compatible with the multiplication map $G \times G \rightarrow G$, i.e., satisfying $\tau(gx \otimes g'y) = gg'\tau(x \otimes y)$. In other words, writing $xy = \tau(x \otimes y)$ for $x, y \in F$, what we have is a $\mathbb{Z}G$ -bilinear product on F such that (a) $\varepsilon(xy) = \varepsilon(x)\varepsilon(y)$ and (b) $\partial(xy) = \partial x \circ y + (-1)^{\deg x} x \circ \partial y$. (For brevity, we will call a $\mathbb{Z}G$ -bilinear product on F satisfying (a) and (b) an *admissible product*.) Such a product induces a k -bilinear product on $F \otimes_G k$, which in turn induces the Pontryagin product on $H_*(G, k)$. As in 3.4–3.6, one verifies that this product makes $H_*(G, k)$ an associative, anti-commutative, graded k -algebra with identity.

In case F is the standard resolution, there is a canonical product on F , called the *shuffle product*. This has its origins in the Eilenberg–Zilber theorem of algebraic topology (cf. MacLane [1963], VIII.8, X.12), and in the present context it takes the following form. Let $\mathcal{S}_n$ be the group of permutations of $\{1, \dots, n\}$, and let $\mathcal{S}_n$ act $\mathbb{Z}G$ -linearly on F_n by

$$\sigma[g_1 | \cdots | g_n] = (-1)^{\text{sgn } \sigma} [g_{\sigma^{-1}(1)} | \cdots | g_{\sigma^{-1}(n)}].$$

If $n = p + q$ then an element $\sigma \in \mathcal{S}_n$ is said to be a (p, q) -*shuffle* if $\sigma(i) < \sigma(j)$ for $1 \leq i < j \leq p$ and for $p + 1 \leq i < j \leq p + q$. [Such a permutation can be viewed as a way of shuffling a deck of p cards with a deck of q cards.] The shuffle product on F is now defined to be the $\mathbb{Z}G$ -bilinear product such that

$$[g_1 | \cdots | g_p] \cdot [g_{p+1} | \cdots | g_{p+q}] = \sum_{\sigma} \sigma[g_1 | \cdots | g_{p+q}],$$

where σ ranges over all (p, q) -shuffles. One can verify that, in addition to being admissible, this product is associative, anti-commutative, and has an identity. Since F is $\mathbb{Z}$ -torsion-free, anti-commutativity implies *strict* anti-commutativity, i.e., $x^2 = 0$ if $\deg x$ is odd. It follows easily that $F \otimes_G k$ is strictly anti-commutative. Consequently:

(5.1) Proposition. *For any abelian group G and commutative ring k , the ring $H_*(G, k)$ is strictly anti-commutative.*

(By contrast, the cohomology ring $H^*(G, k)$ is not strictly anti-commutative in general. An attempt to prove strict anti-commutativity by the method used above fails because the Alexander–Whitney cochain product is only anti-commutative up to homotopy.)

We now give some other examples of resolutions with admissible products.

(5.2) Let G be infinite cyclic and let F be the resolution of I.4.5. Let 1 and x denote the $\mathbb{Z}G$ -basis elements in dimensions 0 and 1, respectively. We set $1 \cdot 1 = 1$, $1 \cdot x = x \cdot 1 = x$, and $x^2 = 0$, and it is trivial to verify admissibility.

(Note that F , as a $\mathbb{Z}G$ -algebra, is the *exterior algebra* $\bigwedge(x)$ with one generator x of degree 1.)

Our next example will use the notion of *divided polynomial algebra*, which is defined as follows: Consider the elements $y^{(i)} = y^i/i!$ ($i \geq 0$) in the polynomial ring $\mathbb{Q}[y]$. We have $y^{(i)}y^{(j)} = (i, j)y^{(i+j)}$, where (i, j) is the binomial coefficient $\binom{i+j}{i} = (i+j)!/i!j!$, so the $\mathbb{Z}$ -submodule of $\mathbb{Q}[y]$ generated by the $y^{(i)}$ is a subring. This ring is denoted $\Gamma(y)$ or $\Gamma_{\mathbb{Z}}(y)$ and is called the divided polynomial ring (over $\mathbb{Z}$) in one variable y . More generally, if R is an arbitrary ring, then the R -algebra $R \otimes \Gamma(y)$ is called the divided polynomial algebra over R in one variable y and is denoted $\Gamma(y)$ or $\Gamma_R(y)$. We will regard $\Gamma(y)$ as a *graded ring*, with $\deg y = 2$.

(5.3) Let G be a finite cyclic group and let F be the resolution I.6.3 with one $\mathbb{Z}G$ -basis element e_i in dimension i , $\partial e_{2i} = Ne_{2i-1}$ ($i > 0$), and $\partial e_{2i-1} = (i-1)e_{2i-2}$. Define a $\mathbb{Z}G$ -bilinear product on F by:

- (i) $e_{2i}e_{2j} = (i, j)e_{2i+2j}$
- (ii) $e_{2i}e_{2j+1} = (i, j)e_{2i+2j+1} = e_{2j+1}e_{2i}$
- (iii) $e_{2i+1}e_{2j+1} = 0$.

Thus F , as a $\mathbb{Z}G$ -algebra, is simply $\bigwedge(e_1) \otimes_{\mathbb{Z}G} \Gamma(e_2)$. [Recall that if A and B are graded algebras over a commutative ring R , then their graded tensor product $A \otimes_R B$ is a graded algebra, with

$$(a \otimes b)(a' \otimes b') = (-1)^{\deg b \cdot \deg a'} aa' \otimes bb']$$

One can check that this product on F is admissible; it is also associative and anti-commutative and has an identity.

Remark. The motivation for the definition above of the product in F is as follows: Let $x = e_1$ and $y = e_2$. Suppose we want our product to be associative and anti-commutative, to have e_0 as identity, and to satisfy (iv) $xe_{2i} = e_{2i+1}$. Let $y^i = c_i e_{2i}$, where c_i is an element of $\mathbb{Z}G$ to be determined. Then $\partial(y^i) = c_i \partial(e_{2i}) = c_i Ne_{2i-1}$. On the other hand, admissibility and (iv) imply that

$$\partial(y^i) = iy^{i-1} \partial y = ic_{i-1} e_{2i-2} Nx = ic_{i-1} Ne_{2i-1}.$$

So we must have $c_i N = ic_{i-1} N$. The most obvious way to satisfy this is to take $c_i = i!$, so that $e_{2i} = y^i/i!$. Formulas (i)–(iii) now follow at once.

(5.4) Suppose F and F' are resolutions with admissible product for G and G' . Then it is easily verified that $F \otimes F'$ is a resolution with admissible product for $G \times G'$, where, as usual,

$$(x \otimes x')(y \otimes y') = (-1)^{\deg x' \cdot \deg y} xy \otimes x'y'.$$

Combining examples 5.2–5.4, one can write down a resolution with admissible product for any finitely generated abelian group G .

Finally, we use 5.4 to relate the cross-product to the Pontryagin product. If F and F' are as in 5.4, there is a chain isomorphism

$$(5.5) \quad (F \otimes_G k) \otimes_k (F' \otimes_{G'} k) \rightarrow (F \otimes F') \otimes_{G \times G'} k,$$

given by $(x \otimes \lambda) \otimes (x' \otimes \lambda') \mapsto (x \otimes x') \otimes \lambda\lambda'$. For $z \in F \otimes_G k$ and $z' \in F' \otimes_{G'} k$ we denote by $z \times z'$ the image of $z \otimes z'$ under this map. As in §2, there is an induced homology cross-product

$$(5.6) \quad H_*(G, k) \otimes_k H_*(G', k) \xrightarrow{\times} H_*(G \times G', k).$$

(5.7) Proposition. *If G and G' are abelian, then the cross-product 5.6 is a k -algebra homomorphism. Moreover, $z \times z' = i_* z \cdot i'_* z'$ for any $z \in H_*(G, k)$ and $z' \in H_*(G', k)$, where $i: G \rightarrow G \times G'$ and $i': G' \rightarrow G \times G'$ are the inclusions.*

PROOF. The first assertion is an easy consequence of 5.4. Since $z \otimes z' = (z \otimes 1) \cdot (1 \otimes z')$, it now follows that $z \times z' = (z \times 1) \cdot (1 \times z')$. I claim that $z \times 1 = i_* z$; indeed, by naturality of the cross-product it suffices to check this for $G' = 1$, in which case it is obvious. Similarly, $1 \times z' = i'_* z'$, whence the proposition. $\square$

We can now state the following Künneth formula:

(5.8) Corollary. *If G and G' are abelian and k is a principal ideal domain, then there is a split-exact sequence*

$$\begin{aligned} 0 &\rightarrow \bigoplus_{p+q=n} H_p(G, k) \otimes_k H_q(G', k) \xrightarrow{\mu} H_n(G \times G', k) \\ &\rightarrow \bigoplus_{p+q=n-1} \text{Tor}_1^k(H_p(G, k), H_q(G', k)) \rightarrow 0, \end{aligned}$$

where $\mu(z \otimes z') = i_* z \cdot i'_* z'$.

PROOF. In view of the isomorphism 5.5 and the Künneth theorem for chain complexes, we have a split-exact sequence as above with μ equal to the cross-product map 5.6. [This is true even if G and G' are non-abelian.] Now apply 5.7. $\square$

EXERCISES

1. Recall that if R is a commutative ring, then $M \otimes_R N$ has an R -module structure for any two R -modules M and N , defined by $r \cdot (m \otimes n) = rm \otimes n = m \otimes rn$. This applies in particular if $R = \mathbb{Z}G$ with G abelian. The resulting tensor product $M \otimes_{\mathbb{Z}G} N$ should not be confused with the tensor product $M \otimes_G N = (M \otimes N)_G$ which we introduced in §III.0 for arbitrary G . (By contrast, the present tensor product $M \otimes_{\mathbb{Z}G} N$, as an abelian group, is equal to $(M \otimes N)_G$ where G acts *anti-diagonally* on $M \otimes N$: $g \cdot (m \otimes n) = g^{-1}m \otimes gn$.)

- (a) If G is abelian, show that there is a Pontryagin product $H_*(G, M) \otimes H_*(G, N) \rightarrow H_*(G, M \otimes_{\mathbb{Z}G} N)$.
- (b) Give an example where $M \otimes_{\mathbb{Z}G} N \not\cong M \otimes_G N$. [Hint: Take G cyclic of order 3 and $M = N = \mathbb{Z}_7$, with non-trivial G -action.]
2. State and prove a result analogous to 5.7, expressing the cohomology cross-product in terms of the cup product.
3. (a) Prove that homology commutes with direct limits. More precisely, let $(G_\alpha)_{\alpha \in D}$ be a direct system of groups, where D is a directed set, and let $G = \varinjlim G_\alpha$. For any G -module M , we have a compatible family of maps $H_*(G_\alpha, M) \rightarrow H_*(G, M)$ ($\alpha \in D$), hence a map $\varphi: \varinjlim H_*(G_\alpha, M) \rightarrow H_*(G, M)$. Prove that φ is an isomorphism. [Hint: Use the standard resolution.]
- (b) Prove 5.1 without using the shuffle product. [Hint: Use (a) to reduce to the case where G is finitely generated.]
4. Let $n = p + q + r$. Define the notion of (p, q, r) -shuffle and prove that the 3-fold shuffle product in the bar resolution is given by

$$[g_1 | \cdots | g_p] \cdot [g_{p+1} | \cdots | g_{p+q}] \cdot [g_{p+q+1} | \cdots | g_{p+q+r}] = \sum_{\sigma} \sigma [g_1 | \cdots | g_{p+q+r}],$$

where σ ranges over the (p, q, r) -shuffles. Generalize.

6 Application: Calculation of the Homology of an Abelian Group

For any abelian group G we have $H_1(G, k) = H_1(G) \otimes k = G \otimes k$. We can therefore construct elements of $H_*(G, k)$ by taking products in the sense of §5 of elements of $G \otimes k$. (If $k = \mathbb{Z}$, for example, the reader has already seen this construction in exercise 1 of §II.3.) Our first goal in this section is to show (at least if k is a principal ideal domain) that there are no relations among these products other than the relations imposed by the fact that $H_*(G, k)$ is a strictly anti-commutative k -algebra. We begin by reviewing the notion of exterior algebra, so that we can state this result precisely.

Let k be an arbitrary commutative ring, let V be a k -module, and let $T^p(V) = V \otimes \cdots \otimes V$ (p copies of V), where $\otimes = \otimes_k$. (By convention, $T^0(V) = k$.) There is an obvious k -bilinear product on $T^*(V)$ given by juxtaposition of tensors, making $T^*(V)$ a graded k -algebra. We define the *exterior algebra* of V , denoted $\bigwedge^*(V)$ or $\bigwedge_k^*(V)$, to be the quotient of $T^*(V)$ by the two-sided ideal generated by the elements $v \otimes v \in T^2(V)$. Explicitly, then, $\bigwedge^p(V)$ is the quotient of $T^p(V)$ by the k -submodule generated by the elements $v_1 \otimes \cdots \otimes v_p$ such that $v_i = v_{i+1}$ for some i . The product in $\bigwedge^*(V)$ is sometimes denoted $x \wedge y$.

We have $\bigwedge^0(V) = k$ and $\bigwedge^1(V) = V$; moreover, $\bigwedge^1(V)$ generates $\bigwedge^*(V)$ as a k -algebra. Since the generators $v \in V = \bigwedge^1(V)$ satisfy $v^2 = 0$ by construction, it follows easily that $\bigwedge^*(V)$ is *strictly anti-commutative*. Moreover, it is immediate from the definition that $\bigwedge^*(V)$ has the following universal property:

(6.1) If A^* is a strictly anti-commutative graded k -algebra, then any k -module map $V \rightarrow A^1$ extends uniquely to a k -algebra map $\bigwedge^*(V) \rightarrow A^*$:

$$\begin{array}{ccc} V & \longrightarrow & A^* \\ \downarrow & \nearrow & \\ \bigwedge^*(V) & & \end{array}$$

Thus $\bigwedge^*(V)$ is the strictly anti-commutative k -algebra “freely generated” by V . The following two properties of the exterior algebra functor will be crucial in our study of homology:

(6.2) $\bigwedge^*(V_1 \oplus V_2) \approx \bigwedge^*(V_1) \otimes \bigwedge^*(V_2)$. More precisely, let $i_1: \bigwedge^*(V_1) \rightarrow \bigwedge^*(V_1 \oplus V_2)$ and $i_2: \bigwedge^*(V_2) \rightarrow \bigwedge^*(V_1 \oplus V_2)$ be induced by the inclusions $V_1 \hookrightarrow V_1 \oplus V_2$ and $V_2 \hookrightarrow V_1 \oplus V_2$, and let $\varphi: \bigwedge^*(V_1) \otimes \bigwedge^*(V_2) \rightarrow \bigwedge^*(V_1 \oplus V_2)$ be defined by $\varphi(x \otimes y) = i_1 x \cdot i_2 y$. Then φ is an isomorphism.

PROOF. There is a k -map $V_1 \oplus V_2 \rightarrow \bigwedge^*(V_1) \otimes \bigwedge^*(V_2)$ given by $v_1 \mapsto v_1 \otimes 1, v_2 \mapsto 1 \otimes v_2$ ($v_i \in V_i$). By 6.1 this extends to an algebra map

$$\bigwedge^*(V_1 \oplus V_2) \rightarrow \bigwedge^*(V_1) \otimes \bigwedge^*(V_2),$$

which is the inverse of φ . □

(6.3) Let $(V_\alpha)_{\alpha \in D}$ be a direct system of k -modules, where D is a directed set. Then $\bigwedge^*(\varinjlim V_\alpha) \approx \varinjlim \bigwedge^*(V_\alpha)$. More precisely, the canonical maps $V_\alpha \rightarrow \varinjlim V_\alpha$ induce a compatible family of maps $\bigwedge^*(V_\alpha) \rightarrow \bigwedge^*(\varinjlim V_\alpha)$ and hence a map $\varphi: \varinjlim \bigwedge^*(V_\alpha) \rightarrow \bigwedge^*(\varinjlim V_\alpha)$; this map φ is an isomorphism.

PROOF. The inclusions $V_\alpha \hookrightarrow \bigwedge^*(V_\alpha)$ induce a k -map $\varinjlim V_\alpha \rightarrow \varinjlim \bigwedge^*(V_\alpha)$. This extends by 6.1 to an algebra map $\bigwedge^*(\varinjlim V_\alpha) \rightarrow \varinjlim \bigwedge^*(V_\alpha)$, which is the inverse of φ . □

Using 6.2 and 6.3, one can give the following concrete description of $\bigwedge^*(V)$ if V is a free k -module: Choose a basis $(x_i)_{i \in I}$ with I simply ordered; then $\bigwedge^p(V)$ has a k -basis consisting of the monomials $x_{i_1} \cdots x_{i_p}$ with $i_1 < \cdots < i_p$. One often writes $\bigwedge^*(V) = \bigwedge^*(x_i)_{i \in I}$.

Returning now to the study of $H_*(G, k)$ for G abelian, the isomorphism $G \otimes k \rightarrow H_1(G, k)$ extends by 6.1 to a k -algebra map $\psi: \bigwedge^*(G \otimes k) \rightarrow H_*(G, k)$, which is an isomorphism in dimensions 0 and 1. It is an obvious but important fact that ψ is a *natural* map of functors of G .

(6.4) Theorem. Assume that k is a principal ideal domain.

- (i) The map $\psi: \bigwedge^*(G \otimes k) \rightarrow H_*(G, k)$ is injective for every abelian group G and is a split injection if G is finitely generated.
- (ii) Suppose that every prime p such that G has p -torsion is invertible in k ; then ψ is an isomorphism.
- (iii) If k has characteristic zero (e.g., if $k = \mathbb{Z}$) then ψ is an isomorphism in dimension 2.

Note that the hypothesis of (ii) holds, in particular, if $k = \mathbb{Q}$, or if $k = \mathbb{Z}_p$ and G is p -torsion-free, or if $k = \mathbb{Z}$ and G is torsion-free.

PROOF OF 6.4. Suppose first that G is cyclic. Then $\bigwedge^p(G \otimes k) = 0$ for $p > 1$, so (i) holds trivially. Under the hypothesis of (ii) we also have $H_p(G, k) = 0$ for $p > 1$ by the computation in §III.1, so ψ is an isomorphism. The same computation shows that $H_2(G, k) = 0$ if $\text{char } k = 0$, so (iii) holds.

Now suppose G is finitely generated and hence a finite direct product of cyclic groups. Arguing by induction on the number of cyclic factors, we may assume that $G = G_1 \times G_2$ and that the theorem is known for G_1 and G_2 . Consider the diagram

$$\begin{array}{ccc}
 \bigwedge^*(G_1 \otimes k) \otimes_k \bigwedge^*(G_2 \otimes k) & \xrightarrow[\cong]{\varphi} & \bigwedge^*(G \otimes k) \\
 \psi(G_1) \otimes \psi(G_2) \downarrow & & \downarrow \psi(G) \\
 H_*(G_1, k) \otimes_k H_*(G_2, k) & \xrightarrow{\mu} & H_*(G, k),
 \end{array}$$

where φ is the isomorphism of 6.2 and μ is the split injection of 5.8. Looking at the definitions of φ and μ and using the naturality of ψ , we see that this diagram commutes. Since $\psi(G_1)$ and $\psi(G_2)$ are split injections by hypothesis, it now follows that $\psi(G)$ is a split injection, whence (i). Now assume that the hypothesis of (ii) holds. Then it also holds for G_1 and G_2 , so $\psi(G_1)$ and $\psi(G_2)$ are isomorphisms. Since $G_i \otimes k$ is a free k -module (with one basis element for every infinite cyclic factor of G_i), it follows that $H_*(G_i, k)$ is k -free. Therefore the Tor term in the Künneth formula vanishes, so μ is an isomorphism and (ii) follows at once. Similarly, μ is always an isomorphism in dimension 2 because the Tor term involves $H_0(-, k) = k$. Assuming now that $\text{char } k = 0$, we know that $\psi(G_1)$ and $\psi(G_2)$ are isomorphisms in dimensions ≤ 2 , whence (iii).

Finally, any group G is the direct limit of its finitely generated subgroups G_α . We therefore have a diagram

$$\begin{array}{ccc}
 \varinjlim \bigwedge^*(G_\alpha \otimes k) & \xrightarrow[\cong]{\varphi} & \bigwedge^*(G \otimes k) \\
 \varinjlim \psi(G_\alpha) \downarrow & & \downarrow \psi(G) \\
 \varinjlim H_*(G_\alpha, k) & \xrightarrow[\cong]{} & H_*(G, k),
 \end{array}$$

where φ is the isomorphism of 6.3 and the unlabelled isomorphism is that of exercise 3a of §5. As above, this diagram commutes because ψ is natural, and the theorem follows at once. $\square$

If G has p -torsion and p is not invertible in k , then the situation is more complicated. For simplicity we will confine ourselves to the case $k = \mathbb{Z}_p$, and even in this case we will be somewhat sketchy. See Cartan [1954/55] for more details and for the calculation of $H_*(G, \mathbb{Z})$.

Recall from 5.3 that if G is a finite cyclic group with generator t then there is a free resolution F of the form $\bigwedge(x) \otimes_{\mathbb{Z}G} \Gamma(y)$, where $\deg x = 1$, $\deg y = 2$, $\partial x = (t - 1) \cdot 1$, and $\partial y = Nx$. If p is a prime dividing $|G|$, then $F \otimes_G \mathbb{Z}_p = \bigwedge(x) \otimes \Gamma(y)$ with $\partial x = 0$ and $\partial y = 0$, where $\bigwedge$, $\otimes$, and Γ are now over $\mathbb{Z}_p$; hence $H_*(G, \mathbb{Z}_p) \approx \bigwedge(x) \otimes \Gamma(y)$. Thus we have, in addition to the exterior algebra part of $H_*(G, \mathbb{Z}_p)$ which we understand via 6.4, a 2-dimensional generator y and its divided powers $y^{(i)}$. This suggests that the homology of an abelian group should have, in addition to its ring structure, a “divided power” structure. We now make this precise.

Let $A = (A_n)_{n \geq 0}$ be a strictly anti-commutative graded ring. By a *system of divided powers* on A we mean a family of functions $A_{2n} \rightarrow A_{2ni}$ for $n > 0$ and $i \geq 0$, denoted $x \mapsto x^{(i)}$, with the following properties:

- (a) $x^{(0)} = 1$, $x^{(1)} = x$.
- (b) $x^{(i)}x^{(j)} = (i, j)x^{(i+j)}$, where $(i, j) = (i + j)!/i!j!$.
- (c) $(x + y)^{(i)} = \sum_{j+k=i} x^{(j)}y^{(k)}$.
- (d) $(xy)^{(i)} = \begin{cases} 0 & \text{if } \deg x \text{ and } \deg y \text{ are odd and } i \geq 2 \\ x^i y^{(i)} & \text{if } \deg x \text{ and } \deg y \text{ are even and } \deg y > 0. \end{cases}$
- (e) $(x^{(i)})^{(j)} = \varepsilon_{ij}x^{(ij)}$ if $i, j > 0$, where

$$\varepsilon_{ij} = (i, i - 1)(2i, i - 1) \cdots ((j - 1)i, i - 1).$$

If A has a differential ∂ (which is always assumed to satisfy $\partial(xy) = \partial x \cdot y + (-1)^{\deg x} x \cdot \partial y$), then we require also:

- (f) $\partial x^{(i)} = x^{(i-1)}\partial x$ for $i > 0$.

Note that this implies that $x^{(i)}$ is a cycle if x is a cycle. But there is not necessarily an induced system of divided powers on H_*A ; see exercise 3 below.

In working with divided powers it is often convenient to introduce the formal power series $e^{tx} = \sum_{i \geq 0} x^{(i)}t^i$, where t is an indeterminate. Formula (c) then takes the form $e^{t(x+y)} = e^{tx}e^{ty}$, so $x \mapsto e^{tx}$ is a homomorphism from the additive group A_{2n} ($n > 0$) to the multiplicative group of formal power series $1 + y_1t + y_2t^2 + \cdots$ with $y_i \in A_{2ni}$.

(6.5) EXAMPLES

1. If A is a strictly anti-commutative graded $\mathbb{Q}$ -algebra, then A admits a system of divided powers with $x^{(i)} = x^i/i!$. Moreover, this is the unique system of divided powers on A ; for (a) and (b) imply that, in any algebra with divided powers, $xx^{(j-1)} = jx^{(j)}$ and hence $x^i = i!x^{(i)}$.

2. Let $\alpha: R \rightarrow S$ be a homomorphism of commutative rings and let A be a (strictly anti-commutative) graded R -algebra with a system of divided powers. Then the S -algebra $S \otimes_R A$ obtained by extension of scalars admits a unique system of divided powers such that $(1 \otimes x)^{(i)} = 1 \otimes x^{(i)}$. Indeed, we must have $(s \otimes x)^{(i)} = s^i \otimes x^{(i)}$ by (d), so uniqueness follows from (c). To prove existence, one can use the universal mapping property of the tensor product to define e^{t^2} for $z \in S \otimes_R A_{2n}$ by $e^{t(s \otimes x)} = \sum_i (s^i \otimes x^{(i)})t^i$; details are left to the reader.

3. The divided polynomial algebra $\Gamma_R(y)$ defined in §5 admits a system of divided powers. Indeed, in view of example 2, it suffices to check this if $R = \mathbb{Z}$, in which case the assertion follows from the fact that $\Gamma(y) \subset \mathbb{Q}[y]$ is closed under the divided power operations [cf. identity (e)]. In exactly the same way we can construct a divided polynomial algebra $\Gamma_R(y_i)_{i \in I}$ in several variables; if I is simply ordered, then this has an R -basis consisting of the “monomials” $y_{i_1}^{(n_1)} \cdots y_{i_k}^{(n_k)}$ with $i_1 < \cdots < i_k$ and $n_j \geq 0$. If V is a free R -module with basis $(y_i)_{i \in I}$, then we set $\Gamma(V) = \Gamma_R(y_i)$; this has V in dimension 2 and has a universal mapping property (in the category of R -algebras with divided powers) analogous to 6.1. It follows, in particular, that $\Gamma(V)$ is well-defined up to canonical isomorphism, independent of the choice of basis.

4. The Pontryagin ring $H_*(G, k)$ of an abelian group G admits a natural system of divided powers. To see this, consider the bar resolution F with the shuffle product. If $x = [g_1 | \cdots | g_{2n}]$ ($n > 0$), then we have (cf. §5, exercise 4)

$$x^i = \sum_{\sigma \in \mathcal{S}\mathcal{H}} \sigma[g_1 | \cdots | g_{2n} | \cdots | g_1 | \cdots | g_{2n}],$$

where the “block” $g_1 | \cdots | g_{2n}$ is repeated i times and $\mathcal{S}\mathcal{H} \subset \mathcal{S}_{2ni}$ is the set of $(2n, \dots, 2n)$ -shuffles. Now there is an obvious embedding $\mathcal{S}_i \hookrightarrow \mathcal{S}_{2ni}$, obtained by letting $\mathcal{S}_i$ permute i sets of $2n$ elements blockwise. Since $\mathcal{S}_i \subset \mathcal{S}_{2ni}$ is a group of even permutations, it is clear that $\mathcal{S}_i$ fixes

$$[g_1 | \cdots | g_{2n} | \cdots | g_1 | \cdots | g_{2n}].$$

Moreover, $\mathcal{S}\mathcal{H}$ is closed under right multiplication by elements of $\mathcal{S}_i$; the above formula for x^i can therefore be written

$$x^i = i! \sum_{\sigma \in \mathcal{S}\mathcal{H}\mathcal{S}_i} \sigma[g_1 | \cdots | g_{2n} | \cdots | g_1 | \cdots | g_{2n}].$$

It follows at once (via identity (c)) that $F \subset \mathbb{Q} \otimes F$ is closed under the divided power operations and hence admits a system of divided powers.

This yields (cf. example 2) a system of divided powers on $k \otimes_G F$, and I claim that there is an induced system of divided powers on $H_*(G, k)$. To prove this, it suffices (cf. exercise 3 below) to show that $z^{(i)}$ is a boundary if $z \in k \otimes_G F_{2n}$ is a boundary. For this purpose we view $k \otimes_G F$ as $(k \otimes F)_G$ and note that $k \otimes F$ is acyclic in positive dimensions (because $\varepsilon: F \rightarrow \mathbb{Z}$ is a homotopy equivalence if we forget the G -action). Now any boundary $z \in k \otimes_G F$ is the image of a boundary $w \in k \otimes F$, and $w^{(i)}$ is a cycle by identity (f). But then $w^{(i)}$ is a boundary by acyclicity, and hence its image $z^{(i)}$ is also a boundary.

Returning now to the study of $H_*(G, \mathbb{Z}_p)$, consider the split-exact universal coefficient sequence

$$0 \rightarrow H_2 G \otimes \mathbb{Z}_p \rightarrow H_2(G, \mathbb{Z}_p) \rightarrow \text{Tor}(H_1 G, \mathbb{Z}_p) \rightarrow 0.$$

We have $H_2 G \approx \bigwedge^2(G)$ by 6.4(iii), and it is easily seen that $\bigwedge^2(G) \otimes \mathbb{Z}_p \approx \bigwedge_{\mathbb{Z}_p}^2(G_p)$, where $G_p = G \otimes \mathbb{Z}_p = G/pG$. Also, $\text{Tor}(H_1 G, \mathbb{Z}_p) = \text{Tor}(G, \mathbb{Z}_p) = {}_p G$, where the latter denotes $\{g \in G: pg = 0\}$. The sequence above therefore takes the form

$$0 \rightarrow \bigwedge^2(G_p) \rightarrow H_2(G, \mathbb{Z}_p) \rightarrow {}_p G \rightarrow 0.$$

Choose a splitting ${}_p G \rightarrow H_2(G, \mathbb{Z}_p)$ of this sequence. (If p is odd, we may use the *canonical splitting* given in exercise 4b below.) This extends to a $\mathbb{Z}_p$ -algebra homomorphism $\varphi: \Gamma({}_p G) \rightarrow H_*(G, \mathbb{Z}_p)$ compatible with divided powers, cf. example 3 above. Combining φ with the map $\psi: \bigwedge(G_p) \rightarrow H_*(G, \mathbb{Z}_p)$ studied in 6.4, we obtain an algebra map $\rho: \bigwedge(G_p) \otimes \Gamma({}_p G) \rightarrow H_*(G, \mathbb{Z}_p)$ given by $\rho(x \otimes y) = \psi(x)\varphi(y)$. We can now state:

(6.6) Theorem. *The map $\rho: \bigwedge(G_p) \otimes \Gamma({}_p G) \rightarrow H_*(G, \mathbb{Z}_p)$ is an isomorphism. It is natural if $p \neq 2$.*

If G is cyclic, this follows easily from our earlier computations. The general case can now be deduced by using the Künneth formula and direct limits as in the proof of 6.4. See Cartan [1954/55] for more details.

EXERCISES

1. If G is an abelian group (written additively) and $n \in \mathbb{Z}$, compute the endomorphism of $H_*(G, \mathbb{Q})$ induced by the endomorphism $g \mapsto ng$ of G .
2. Let A and B be strictly anti-commutative graded k -algebras. Show that $A \otimes_k B$ is the sum of A and B in the category of strictly anti-commutative graded k -algebras, via the maps $a \mapsto a \otimes 1$, $b \mapsto 1 \otimes b$ ($a \in A$, $b \in B$). Thus 6.2 can be interpreted as saying that the exterior algebra functor preserves sums.

3. Let A be a strictly anti-commutative graded ring with a differential ∂ and a system of divided powers.
- (a) Suppose that $x^{(i)}$ is a boundary whenever x is a boundary. Show that there is an induced system of divided powers on $H_* A$.
- (b) Give an example to show that the hypothesis of (a) need not hold.
4. (a) Show that the injection $\psi: \bigwedge^2(G \otimes k) \rightarrow H_2(G, k)$ of 6.4 takes $(g \otimes 1) \wedge (h \otimes 1)$ to the class of $[g|h] - [h|g]$.
- (b) If 2 is invertible in k , show that the map $C_2(G, k) \rightarrow \bigwedge^2(G \otimes k)$ given by $[g|h] \mapsto (g \otimes 1) \wedge (h \otimes 1)/2$ induces a map $H_2(G, k) \rightarrow \bigwedge^2(G \otimes k)$ which is a left inverse of ψ .
5. Let G be abelian and let A be a G -module with trivial G -action. In view of the isomorphism $H_2 G \approx \bigwedge^2 G$, the universal coefficient theorem (cf. exercise 3 of §III.1) gives us a split exact sequence

$$0 \rightarrow \text{Ext}(G, A) \rightarrow H^2(G, A) \xrightarrow{\theta} \text{Hom}(\bigwedge^2 G, A) \rightarrow 0.$$

Deduce from exercise 4a that θ coincides with the map called θ in exercise 8 of §IV.3, hence the latter is a (split) surjection. Thus every alternating map comes from a 2-cocycle. (A non-cohomological proof of this surprising fact can be found in Hughes [1951], proof of Theorem 2.) Incidentally, it also follows that $\text{Ext}(G, A) \approx \mathcal{E}_{ab}(G, A)$. [More generally, it is known for any ring R that $\text{Ext}_R^1(M, N)$ is isomorphic to the set of equivalence classes of R -module extensions of M by N , whence the name “Ext.” A proof of this can be found in almost any book on homological algebra.]

CHAPTER VI

Cohomology Theory of Finite Groups

1 Introduction

Homology and cohomology are usually thought of as dual to one another. We have seen in Chapter III, for example, that homology has a number of formal properties and that cohomology has “dual” properties. If G is finite, however, then homology and cohomology seem to have *similar* properties rather than dual ones. For example, since every subgroup H of a finite group G has finite index, we have restriction and corestriction maps for *arbitrary* subgroups, in both homology and cohomology. For another example, the distinction between induced modules and co-induced modules disappears, so we have a single class $\mathcal{J}$ of G -modules (namely, the induced modules $\mathbb{Z}G \otimes A$) with the following properties: (a) Every $M \in \mathcal{J}$ is acyclic for both homology and cohomology. (b) For every G -module M there is a module $\bar{M} \in \mathcal{J}$ such that M is a quotient of $\bar{M}$ and M can be embedded in $\bar{M}$.

Tate discovered an ingenious way to exploit these similarities between H_* and H^* for G finite. Namely, he showed that there is a quotient $\tilde{H}^0$ of H^0 and a sub-functor $\tilde{H}_0 \subseteq H_0$ such that the functors $\dots, H_2, H_1, \tilde{H}_0, \tilde{H}^0, H^1, H^2, \dots$ fit together to form a “cohomology theory” $\hat{H}^*$, involving functors $\hat{H}^i$ for all $i \in \mathbb{Z}$:

$$\begin{array}{ccccccc}
 \dots & H_2 & & H_1 & & \tilde{H}_0 & & \tilde{H}^0 & & H^1 & & H^2 \dots \\
 & \parallel & & \parallel & & \parallel & & \parallel & & \parallel & & \parallel \\
 \dots & \hat{H}^{-3} & & \hat{H}^{-2} & & \hat{H}^{-1} & & \hat{H}^0 & & \hat{H}^1 & & \hat{H}^2 \dots
 \end{array}$$

The purpose of this chapter is to develop this *Tate cohomology theory* and to illustrate its usefulness by discussing (a) the Nakayama–Rim theory of co-

homologically trivial modules (§8) and (b) the theory of groups with periodic cohomology (§9).

2 Relative Homological Algebra

We indicated in exercise 2 of §I.7 that the fundamental lemma I.7.4 and its corollaries could be generalized in various ways. In particular, we will need in this chapter a “relative-dual” version of those results, which we work out in the present section.

Throughout the section, G will be a fixed group (not necessarily finite) and H a fixed subgroup. For the purposes of the present chapter, the case of interest will be that where G is finite and $H = \{1\}$. But the general case will be needed later, in Chapter X.

An injection $i: M' \hookrightarrow M$ of G -modules will be called *admissible* if it is a split injection when regarded as an injection of H -modules, i.e., if there is an H -map $\pi: M \rightarrow M'$ such that $\pi i = \text{id}_{M'}$. An exact sequence $M' \xrightarrow{i} M \xrightarrow{j} M''$ will be called *admissible* if the inclusion $\text{im } j \hookrightarrow M''$ is admissible. An acyclic chain complex C of G -modules will be called *admissible* if each exact sequence $C_{i+1} \rightarrow C_i \rightarrow C_{i-1}$ is admissible; in view of I.0.3, this is equivalent to saying that C is contractible when regarded as a complex of H -modules. Finally, a G -module Q is *relatively injective* if it satisfies the following equivalent conditions:

(i) Every mapping problem

$$\begin{array}{ccccc} M' & \longrightarrow & M & \longrightarrow & M'' \\ & \searrow & \downarrow & \swarrow & \\ & 0 & & & \\ & & Q & & \end{array}$$

with admissible exact row can be solved.

(ii) Every mapping problem

$$\begin{array}{ccc} M' & \xhookrightarrow{i} & M \\ \downarrow & & \swarrow \\ & & Q \end{array}$$

with i an admissible injection can be solved.

(iii) The contravariant functor $\text{Hom}_G(-, Q)$ takes admissible injections of G -modules to surjections of abelian groups.

In particular, every injective module is certainly relatively injective. But there are many relative injectives which are not injective:

(2.1) Proposition. *For any H -module N , the G -module $\text{Coind}_H^G N$ is relatively injective.*

PROOF. By the universal property of co-induction (III.3.6), we have

$$\text{Hom}_G(-, \text{Coind}_H^G N) \approx \text{Hom}_H(\text{Res}_H^G(-), N).$$

The functor on the right clearly takes any admissible injection of G -modules to a surjection of abelian groups. $\square$

(2.2) Corollary. *For any G -module M there is a canonical admissible injection $M \hookrightarrow \bar{M}$, where $\bar{M}$ is relatively injective. If $(G:H) < \infty$, then this construction has the following properties: (a) If M is free (resp. projective) as a $\mathbb{Z}H$ -module, then $\bar{M}$ is free (resp. projective) over $\mathbb{Z}G$. (b) If M is finitely generated as a G -module, then so is $\bar{M}$.*

PROOF. Take $\bar{M} = \text{Coind}_H^G \text{Res}_H^G M$ and use the canonical H -split injection $M \hookrightarrow \bar{M}$ of III.3.7. If $(G:H) < \infty$, then we have $\text{Coind}_H^G(-) \approx \text{Ind}_H^G(-)$, and (a) and (b) follow easily. $\square$

(2.3) Corollary. *Suppose $(G:H) < \infty$. Then any $\mathbb{Z}G$ -projective module is relatively injective.*

PROOF. It is easy to see that a direct summand of a relative injective is relatively injective, so it suffices to consider free modules. Now if F is a free $\mathbb{Z}G$ -module, then clearly $F \approx \mathbb{Z}G \otimes_{\mathbb{Z}H} F' = \text{Ind}_H^G F'$, where F' is a free $\mathbb{Z}H$ -module of the same rank. But $\text{Ind}_H^G F' \approx \text{Coind}_H^G F'$, which is relatively injective by 2.1. $\square$

We now record the “relative dual” of Lemma I.7.4, in the form in which we will need it:

(2.4) Proposition. *Let C and C' be chain complexes of G -modules and let r be an integer. Suppose that C'_i is relatively injective for $i < r$ and that $C_{i+1} \rightarrow C_i \rightarrow C_{i-1}$ is exact and admissible for $i \leq r$.*

- (a) *Any family $(f_i: C_i \rightarrow C'_i)_{i \geq r}$ of maps commuting with boundary operators extends to a chain map $C \rightarrow C'$.*
- (b) *Let $f, g: C \rightarrow C'$ be chain maps and let $(h_i: C_i \rightarrow C'_{i+1})_{i \geq r-1}$ be a family of maps such that $\partial'_{i+1} h_i + h_{i-1} \partial_i = f_i - g_i$ for $i \geq r$. Then $(h_i)_{i \geq r-1}$ extends to a homotopy from f to g .*

The proof is virtually identical to that of I.7.4, except that all arrows are reversed. $\square$

By a *relative injective resolution* of a G -module M we mean a non-negative cochain complex Q of relative injectives, together with a weak equivalence $\eta: M \rightarrow Q$ such that the augmented complex

$$0 \rightarrow M \rightarrow Q^0 \rightarrow Q^1 \rightarrow \dots$$

is admissible. As an immediate corollary of 2.4, we have:

(2.5) Corollary. *Any two relative injective resolutions of M are canonically homotopy equivalent.*

It is clear that relative injective resolutions exist for any M ; indeed, we can take $\eta: M \rightarrow Q^0$ to be the canonical admissible injection of 2.2, then apply 2.2 again to get an admissible injection $\text{coker } \eta \hookrightarrow Q^1$, etc. Moreover, if $(G: H) < \infty$ and M is projective as a $\mathbb{Z}H$ -module, then I claim that the modules Q^n which occur in this resolution will be projective over $\mathbb{Z}G$. This is clear for Q^0 , by 2.2a. Since η is H -split, it follows that $\text{coker } \eta$ is projective as a $\mathbb{Z}H$ -module, hence 2.2 implies that Q^1 is projective over $\mathbb{Z}G$. Continuing in this way, one proves the claim. Similarly, if M is finitely generated, then so is each Q^n . Summarizing, we have shown:

(2.6) Proposition. *Suppose $(G: H) < \infty$. If M is a $\mathbb{Z}G$ -module which is projective (resp. finitely generated and projective) as a $\mathbb{Z}H$ -module, then M admits a relative injective resolution $\eta: M \rightarrow Q$ such that each Q^n is a projective (resp. finitely generated projective) $\mathbb{Z}G$ -module.*

EXERCISE

Show that the resolution Q obtained in the proof of 2.6 is a complex of free $\mathbb{Z}G$ -modules if M is free as a $\mathbb{Z}H$ -module.

3 Complete Resolutions

We now specialize the relative homological algebra of §2 to the case where G is finite and $H = \{1\}$. As a consequence of 2.6 we obtain the perhaps surprising result that the G -module $\mathbb{Z}$ admits a “backwards” resolution $0 \rightarrow \mathbb{Z} \rightarrow Q^0 \rightarrow Q^1 \rightarrow \dots$, where each Q^i is finitely generated and projective. [This becomes less surprising, however, when one realizes that such a backwards resolution can be obtained by taking the *dual* of an ordinary projective resolution of finite type; cf. 3.5 below.] Setting $F_i = Q^{-i-1}$, this takes the form

$$(3.1) \quad 0 \rightarrow \mathbb{Z} \xrightarrow{\eta} F_{-1} \rightarrow F_{-2} \rightarrow \dots$$

Now splice 3.1 onto an ordinary projective resolution $\varepsilon: (F_n)_{n \geq 0} \rightarrow \mathbb{Z}$, to obtain an acyclic complex

$$\begin{array}{ccccccc} \cdots & \longrightarrow & F_2 & \longrightarrow & F_1 & \longrightarrow & F_0 & \longrightarrow & F_{-1} & \longrightarrow & F_{-2} & \longrightarrow & \cdots \\ & & & & & & \searrow \varepsilon & & \nearrow \eta & & & & \\ & & & & & & & \mathbb{Z} & & & & & \end{array}$$

A complex of this type is called a complete resolution for G . More precisely, a *complete resolution* is an acyclic complex $F = (F_i)_{i \in \mathbb{Z}}$ of projective $\mathbb{Z}G$ -modules, together with a map $\varepsilon: F_0 \rightarrow \mathbb{Z}$ such that $\varepsilon: F_+ \rightarrow \mathbb{Z}$ is a resolution in the usual sense, where $F_+ = (F_i)_{i \geq 0}$. It follows from the definition that $\partial: F_0 \rightarrow F_{-1}$ factors uniquely as $\partial = \eta\varepsilon$, where $\eta: \mathbb{Z} \rightarrow F_{-1}$ is a monomorphism. It also follows that the resulting complex 3.1 is a relative injective resolution of $\mathbb{Z}$. In fact, we know from 2.3 that each F_i is relatively injective, and admissibility follows from exercise 3b of §I.8, which we repeat here:

(3.2) Lemma. *Any acyclic chain complex C of free abelian groups is contractible.*

PROOF. The abelian group Z_n of n -cycles is free, being a subgroup of a free abelian group, so the sequence $0 \rightarrow Z_{n+1} \rightarrow C_{n+1} \rightarrow Z_n \rightarrow 0$ splits. $\square$

We do not require in the definition of complete resolution that F be of finite type, i.e., that each F_i be finitely generated, but we saw at the beginning of this section that there do exist complete resolutions of finite type.

As with ordinary resolutions, we may view the map ε in a complete resolution as a chain map $F \rightarrow \mathbb{Z}$. (Note, however, that it is *not* a weak equivalence.) Given complete resolutions $\varepsilon: F \rightarrow \mathbb{Z}$ and $\varepsilon': F' \rightarrow \mathbb{Z}$, a chain map $\tau: F \rightarrow F'$ is *augmentation-preserving* if $\varepsilon'\tau = \varepsilon$.

(3.3) Proposition. *If $\varepsilon: F \rightarrow \mathbb{Z}$ and $\varepsilon': F' \rightarrow \mathbb{Z}$ are complete resolutions, then there exists a unique homotopy class of augmentation-preserving maps from F to F' . These maps are homotopy equivalences.*

PROOF. By I.7.4 we can find an augmentation-preserving chain map $\tau_+: F_+ \rightarrow F'_+$. Since F is acyclic and admissible (by 3.2) and F' is dimension-wise relatively injective, τ_+ can be extended to negative dimensions by 2.4. Similarly, given two augmentation-preserving maps $\tau, \tau': F \rightarrow F'$, I.7.4 gives us a homotopy between τ_+ and τ'_+ , which can then be extended to negative dimensions by 2.4. It is clear from the uniqueness that any map of complete resolutions is a homotopy equivalence. $\square$

Finally, we want to show that the negative part of a complete resolution of finite type is the dual of an ordinary projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$. Recall that a G -module M has a dual $M^* = \text{Hom}_G(M, \mathbb{Z}G)$, which we studied

in I.8.3 for finitely generated projective modules. [Note that M^* is naturally a *right* G -module if M is a left G -module, but we can convert it to a left module as in §III.0. Thus $(gu)(m) = u(m)g^{-1}$ for $g \in G, u \in M^*, m \in M$.] On the other hand, one can also consider the dual $\text{Hom}(M, \mathbb{Z})$ of M as a $\mathbb{Z}$ -module. This is a G -module via the diagonal action (cf. §III.0), where G acts trivially on $\mathbb{Z}$. Thus $(gu)(m) = u(g^{-1}m)$ for $g \in G, u \in \text{Hom}(M, \mathbb{Z}), m \in M$. It turns out that $\text{Hom}(M, \mathbb{Z})$ coincides with the dual M^* when G is finite:

(3.4) Proposition. *For any finite group G and any (left) G -module M , there is a G -module isomorphism*

$$\psi : \text{Hom}(M, \mathbb{Z}) \rightarrow M^* = \text{Hom}_G(M, \mathbb{Z}G),$$

given by $\psi(u)(m) = \sum_{g \in G} u(g^{-1}m)g$ for $u \in \text{Hom}(M, \mathbb{Z}), m \in M$.

PROOF. One can simply verify this by straightforward definition-checking, but in fact it follows from things we have already done. Namely, $\mathbb{Z}G$ is the induced module $\mathbb{Z}G \otimes \mathbb{Z}$, so it is isomorphic to the co-induced module $\text{Hom}(\mathbb{Z}G, \mathbb{Z})$ by III.5.9. The universal property III.3.6 of co-induction therefore implies that $\text{Hom}_G(M, \mathbb{Z}G) \approx \text{Hom}(M, \mathbb{Z})$ (as abelian groups), and an examination of the proofs of III.3.6 and III.5.9 yields the specific isomorphism ψ . It is easy to verify that ψ is compatible with the G -action. $\square$

As an application of this, we will give a concrete interpretation of the “backwards projective resolution” $\mathbb{Z} \rightarrow Q$ discussed at the beginning of this section. By the *dual* of a chain complex F (over an arbitrary ring R) we mean the complex $\bar{F} = \mathcal{H}om_R(F, R)$. Thus $\bar{F}^n = \bar{F}_{-n} = \text{Hom}_R(F_n, R) = (F_n)^*$.

(3.5) Proposition. *If $\varepsilon: P \rightarrow \mathbb{Z}$ is a finite type projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$ (G finite), then $\varepsilon^*: \mathbb{Z}^* = \mathbb{Z} \rightarrow \bar{P}$ is a backwards projective resolution; moreover, up to isomorphism every finite type backwards projective resolution is obtained in this way. Consequently, any finite type complete resolution is obtained from two ordinary finite type projective resolutions $P' \rightarrow \mathbb{Z}$ and $P \rightarrow \mathbb{Z}$ by splicing together P' and $\bar{\Sigma}P$, where ΣP is the suspension of P (§I.0).*

PROOF. The augmented chain complex associated to $\varepsilon: P \rightarrow \mathbb{Z}$ is contractible as a complex of abelian groups (by 3.2, for instance). It therefore remains contractible when the duality functor $(\)^* \approx \text{Hom}(_, \mathbb{Z})$ is applied, so $\varepsilon^*: \mathbb{Z}^* = \mathbb{Z} \rightarrow \bar{P}$ is a backwards projective resolution. This proves the first assertion. Similarly, given any finite type backwards projective resolution $\mathbb{Z} \rightarrow Q$, its dual $\bar{Q} \rightarrow \mathbb{Z}$ is an ordinary projective resolution, and Q can be identified with the dual of $\bar{Q}$ by I.8.3d. The remaining assertion is clear. $\square$

EXERCISE

If G acts freely on S^{2k-1} as in §I.6, show that there is a complete resolution F for G which is periodic of period $2k$. Write this out explicitly if G is finite cyclic.

4 Definition of $\hat{H}^*$

The *Tate cohomology* of a finite group G with coefficients in a G -module M is defined by

$$\hat{H}^i(G, M) = H^i(\mathcal{H}om_G(F, M))$$

for all $i \in \mathbb{Z}$, where F is a complete resolution for G . By 3.3, H^i is well-defined up to canonical isomorphism.

Let $F_+ = (F_i)_{i \geq 0}$ and $F_- = (F_i)_{i < 0}$, and let C^+ (resp. C^-) be the complex $\mathcal{H}om_G(F_+, M)$ (resp. $\mathcal{H}om_G(F_-, M)$). Then we have (cf. proof of II.5.1)

$$\hat{H}^i(G, M) = \begin{cases} H^i(C^+) & i > 0 \\ H^i(C^-) & i < -1 \end{cases}$$

and there is an exact sequence

$$0 \rightarrow \hat{H}^{-1}(G, M) \rightarrow H^{-1}(C^-) \xrightarrow{\alpha} H^0(C^+) \rightarrow \hat{H}^0(G, M) \rightarrow 0,$$

where α is induced by the coboundary operator $\delta: C^{-1} \rightarrow C^0$. More precisely, α is determined by the diagram

$$(*) \quad \begin{array}{ccc} C^{-1} & \xrightarrow{\delta} & C^0 \\ \downarrow & & \uparrow \\ H^{-1}(C^-) & \xrightarrow{\alpha} & H^0(C^+). \end{array}$$

Now $\varepsilon: F_+ \rightarrow \mathbb{Z}$ is a projective resolution of $\mathbb{Z}$, so $H^i(C^+) \approx H^i(G, M)$. And if we assume (as we may) that F is of finite type, then 3.5 implies that $F_- = \overline{\Sigma P}$ for some projective resolution $P \rightarrow \mathbb{Z}$ of finite type. The duality isomorphism I.8.3c now yields

$$C^- = \mathcal{H}om_G(F_-, M) = \mathcal{H}om_G(\overline{\Sigma P}, M) \approx \Sigma P \otimes_G M,$$

so $H^i(C^-) \approx H_{-i}(\Sigma P \otimes_G M) = H_{-i-1}(P \otimes_G M) = H_{-i-1}(G, M)$. We therefore have

$$\hat{H}^i(G, M) = \begin{cases} H^i(G, M) & i > 0 \\ H_{-i-1}(G, M) & i < -1 \end{cases}$$

and there is an exact sequence

$$0 \rightarrow \hat{H}^{-1}(G, M) \rightarrow H_0(G, M) \xrightarrow{\alpha} H^0(G, M) \rightarrow \hat{H}^0(G, M) \rightarrow 0.$$

I claim that α is the norm map $\bar{N}: M_G \rightarrow M^G$ defined in §III.1. To prove this, it is convenient to assume that the projective resolutions F_+ and P above both have $\mathbb{Z}G$ in dimension 0 and both start with the canonical augmentation $\mathbb{Z}G \rightarrow \mathbb{Z}$. [We can certainly assume this, since the resolutions can be taken to

be arbitrary finite type projective resolutions.] Then $\eta = \varepsilon^*: \mathbb{Z} \rightarrow \mathbb{Z}G$ is given by $\eta(1) = \sum_{g \in G} g$. Since $\text{Hom}_G(\mathbb{Z}G, M) = M$, the diagram (*) becomes

$$\begin{array}{ccc} M & \xrightarrow{N} & M \\ \downarrow & & \uparrow \\ H_0(G, M) & \xrightarrow{\alpha} & H^0(G, M), \end{array}$$

and it is easy to check that the vertical maps are the canonical maps $M \rightarrow M_G$ and $M^G \hookrightarrow M$. This proves the claim.

We have now proved

$$\hat{H}^{-1}(G, M) = \ker \bar{N} \subseteq H_0(G, M)$$

and

$$\hat{H}^0(G, M) = \text{coker } \bar{N} \leftarrow H^0(G, M).$$

For example, $\hat{H}^{-1}(G, \mathbb{Z}) = 0$ and $\hat{H}^0(G, \mathbb{Z}) = \mathbb{Z}/|G| \cdot \mathbb{Z}$.

We can summarize the results above by means of the following diagram:

$$\begin{array}{ccccccc} & & & & H^0 & H^1 & H^2 \dots \\ & & & & \downarrow & \parallel & \parallel \\ \dots \hat{H}^{-3} & \hat{H}^{-2} & \hat{H}^{-1} & & \hat{H}^0 & \hat{H}^1 & \hat{H}^2 \dots \\ & \parallel & \parallel & & \parallel & \parallel & \parallel \\ \dots H_2 & H_1 & H_0 & & & & \end{array}$$

(A diagonal arrow labeled $\bar{N}$ points from H_0 to H^0)

One can also define *Tate homology groups*, by $\hat{H}_*(G, M) = H_*(F \otimes_G M)$, where F is a complete resolution. Arguments analogous to those above show that

$$\hat{H}_i = \begin{cases} H_i & i > 0 \\ \ker \bar{N} & i = 0 \\ \text{coker } \bar{N} & i = -1 \\ H^{-i-1} & i < -1. \end{cases}$$

In other words, $\hat{H}_i = \hat{H}^{-i-1}$. In view of this, it may seem pointless to introduce $\hat{H}_*$ since it, like $\hat{H}^*$, just consists of the functors H_i and H^i for $i > 0$ together with modifications of H^0 and H_0 . As we will see in §7, however, the “equality” $\hat{H}_i = \hat{H}^{-i-1}$ should really be viewed as a duality theorem, and, as such, it has important consequences.

5 Properties of $\hat{H}^*$

Because Tate cohomology was defined in terms of resolutions, it is easy to prove that many of the formal properties of H^* hold also for $\hat{H}^*$. For example, one proves as in III.6.1(ii'):

(5.1) A short exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ of G -modules gives rise to a long exact sequence

$$\cdots \xrightarrow{\delta} \hat{H}^i(G, M') \rightarrow \hat{H}^i(G, M) \rightarrow \hat{H}^i(G, M'') \xrightarrow{\delta} \hat{H}^{i+1}(G, M') \rightarrow \cdots$$

Similarly, the proof of Shapiro's lemma (III.6.2) goes through in the present context, since a complete resolution for G can also be regarded as a complete resolution for any $H \subseteq G$. Since co-induction is the same as induction (because G is finite), Shapiro's lemma takes the form:

(5.2) If $H \subseteq G$ and M is an H -module, then $\hat{H}^*(H, M) \approx \hat{H}^*(G, \mathbb{Z}G \otimes_{\mathbb{Z}H} M)$.

Taking $H = \{1\}$ and noting that $\hat{H}^*(\{1\}, -) = 0$, we obtain:

(5.3) $\hat{H}^*(G, \mathbb{Z}G \otimes A) = 0$ for any abelian group A . Consequently, each $\hat{H}^i$ is both effaceable and co-effaceable.

Note that 5.1 and 5.3 allow one to use dimension-shifting. Namely, given a G -module M we can find G -modules K and C (as in III.7.1 and III.7.2) such that:

(5.4) $\hat{H}^i(G, M) \approx \hat{H}^{i+1}(G, K)$ and $\hat{H}^i(G, M) \approx \hat{H}^{i-1}(G, C)$ for all $i \in \mathbb{Z}$.

This shows, at least heuristically, that the Tate cohomology theory is completely determined by any one of the functors $\hat{H}^i$. The interested reader can make this statement precise, as in III.7.3 and III.7.5.

Given $H \subseteq G$, a complete resolution F for G , and a G -module M , we have cochain maps $\mathcal{H}om_G(F, M) \hookrightarrow \mathcal{H}om_H(F, M)$ and $\mathcal{H}om_H(F, M) \rightarrow \mathcal{H}om_G(F, M)$, where the second map is a transfer map, defined as in definition (C) of §III.9. Consequently:

(5.5) For any $H \subseteq G$ and any G -module M , one has a restriction map $\hat{H}^*(G, M) \rightarrow \hat{H}^*(H, M)$ and a corestriction map $\hat{H}^*(H, M) \rightarrow \hat{H}^*(G, M)$. These maps have formal properties analogous to those stated in III.9.5.

It follows, in particular, that the analogues of III.10.2 and III.10.3 hold for $\hat{H}^*$.

Finally, we will show that the theory of cup products extends to $\hat{H}^*$:

(5.6) There is a cup product $\hat{H}^p(G, M) \otimes \hat{H}^q(G, N) \rightarrow \hat{H}^{p+q}(G, M \otimes N)$, with formal properties analogous to 3.1–3.8 of Chapter V.

(Note: To state the analogue of V.3.7, one has to assume that $\alpha: H \rightarrow G$ is an inclusion, since that is the only case where we have a map $\alpha^*: \hat{H}^*(G) \rightarrow \hat{H}^*(H)$.)

In particular, $\hat{H}^*(G, \mathbb{Z})$ is an anti-commutative graded ring with identity element $1 \in \mathbb{Z}/|G| \cdot \mathbb{Z} = \hat{H}^0(G, \mathbb{Z})$, and $\hat{H}^*(G, M)$ is a module over $\hat{H}^*(G, \mathbb{Z})$ for any M .

The construction of the cup product requires some work. Recall that we based our treatment of cup products in ordinary cohomology on the fact that the tensor product of resolutions is a resolution. On the cochain level, then, the cup product was simply the map

$$(5.7) \quad \mathcal{H}om_G(F, M) \otimes \mathcal{H}om_G(F, N) \rightarrow \mathcal{H}om_G(F \otimes F, M \otimes N)$$

given by tensor product of graded maps. We pointed out that one could also define the cup product in terms of a single resolution F by choosing a diagonal approximation $\Delta: F \rightarrow F \otimes F$ and composing the product above with

$$\mathcal{H}om_G(\Delta, M \otimes N): \mathcal{H}om_G(F \otimes F, M \otimes N) \rightarrow \mathcal{H}om_G(F, M \otimes N).$$

Note that, from this point of view, the cup product

$$\mathcal{H}om_G(F, M)^p \otimes \mathcal{H}om_G(F, N)^q \rightarrow \mathcal{H}om_G(F, M \otimes N)^{p+q}$$

depends only on the (p, q) -component $\Delta_{pq}: F_{p+q} \rightarrow F_p \otimes F_q$ of Δ .

Now suppose F is a complete resolution. The first difficulty in trying to imitate the procedure above is that $F \otimes F$ does not appear to be a complete resolution. In particular, $(F \otimes F)_+$ is *not* equal to the resolution $F_+ \otimes F_+$. Consequently, a definition of the form (5.7) would not in any obvious way induce a cohomology product $\hat{H}^p \otimes \hat{H}^q \rightarrow \hat{H}^{p+q}$, so a diagonal approximation now appears to be crucial, rather than a mere convenience. Secondly, a moment's thought shows that $F \otimes F$ is not really the appropriate target for a diagonal approximation. Indeed, for any $n \in \mathbb{Z}$ there are infinitely many (p, q) such that $p + q = n$, and dimension-shifting considerations suggest that the corresponding cup products should all be non-trivial. Thus Δ should have a non-trivial component Δ_{pq} for all (p, q) , so the target of Δ should be the graded module which in dimension n is $\prod_{p+q=n} F_p \otimes F_q$ rather than $\bigoplus_{p+q=n} F_p \otimes F_q$. This discussion motivates the following definitions.

If C and C' are graded modules, their *completed tensor product* $C \hat{\otimes} C'$ is defined by

$$(C \hat{\otimes} C')_n = \prod_{p+q=n} C_p \otimes C'_q.$$

Given two other graded modules D, D' and maps $u: C \rightarrow D$ of degree r and $v: C' \rightarrow D'$ of degree s , there is a map $u \hat{\otimes} v: C \hat{\otimes} C' \rightarrow D \hat{\otimes} D'$ of degree $r + s$ defined by

$$(u \hat{\otimes} v)_n = \prod_{p+q=n} (-1)^{ps} u_p \otimes v_q: \prod_{p+q=n} C_p \otimes C'_q \rightarrow \prod_{p+q=n} D_{p+r} \otimes D'_{q+s}.$$

Now let $\varepsilon: F \rightarrow \mathbb{Z}$ be a complete resolution, and let d be the differential in F . Then $F \hat{\otimes} F$ has the “partial differentials” $d \hat{\otimes} \text{id}_F$ and $\text{id}_F \hat{\otimes} d$; these are of square zero and anti-commute, since $(d \hat{\otimes} \text{id}_F)(\text{id}_F \hat{\otimes} d) = d \hat{\otimes} d$ whereas $(\text{id}_F \hat{\otimes} d)(d \hat{\otimes} \text{id}_F) = -d \hat{\otimes} d$ (cf. exercise 8 of §I.0). So the “total differential” $\partial = d \hat{\otimes} \text{id}_F + \text{id}_F \hat{\otimes} d$ is of square zero and makes $F \hat{\otimes} F$ a chain complex. Moreover, we have the “augmentation” $\varepsilon \hat{\otimes} \varepsilon: F \hat{\otimes} F \rightarrow \mathbb{Z} \hat{\otimes} \mathbb{Z} = \mathbb{Z}$. By a *complete diagonal approximation* we mean an augmentation-preserving chain map $\Delta: F \rightarrow F \hat{\otimes} F$. It is by no means obvious that complete diagonal approximations exist, but we will prove below that they do. Accepting this for the moment, we can define a cochain cup product

$$\mathcal{H}om_G(F, M) \otimes \mathcal{H}om_G(F, N) \rightarrow \mathcal{H}om_G(F, M \otimes N)$$

by $u \cup v = (u \hat{\otimes} v) \circ \Delta$.

One verifies the usual coboundary formula

$$\delta(u \cup v) = \delta u \cup v + (-1)^p u \cup \delta v,$$

and it follows that there is an induced product $\hat{H}^p(G, M) \otimes \hat{H}^q(G, N) \rightarrow \hat{H}^{p+q}(G, M \otimes N)$. It is immediate that this product is natural with respect to coefficient homomorphisms (as in V.3.2), and one proves exactly as in V.3.3 and V.3.3' that it is compatible with connecting homomorphisms δ in long exact cohomology sequences. Moreover, the fact that Δ is augmentation-preserving allows us to calculate the cup product $\hat{H}^0 \otimes \hat{H}^0 \rightarrow \hat{H}^0$, and we find as in V.3.1 that it is induced by the obvious map $M^G \otimes N^G \rightarrow (M \otimes N)^G$. [This makes sense because $\hat{H}^0(G, -)$ is a quotient of $(-)^G$.]

We can now use dimension-shifting to prove that the cup product is independent of the choice of F and Δ . More precisely:

(5.8) Lemma. *There is at most one cup product on $\hat{H}^*(G, -)$ satisfying the analogues of V.3.1, V.3.3, and V.3.3'.*

PROOF. Given a module M , let $\bar{M}$ be the induced module $\mathbb{Z}G \otimes M$ and let $0 \rightarrow K \rightarrow \bar{M} \rightarrow M \rightarrow 0$ be the canonical $\mathbb{Z}$ -split exact sequence. For any G -module N the sequence $0 \rightarrow K \otimes N \rightarrow \bar{M} \otimes N \rightarrow M \otimes N \rightarrow 0$ is exact, and the module $\bar{M} \otimes N$ is induced (cf. §III.5, exercise 2a). We therefore have dimension-shifting isomorphisms

$$\delta: \hat{H}^i(G, M) \xrightarrow{\sim} \hat{H}^{i+1}(G, K) \quad \text{and} \quad \delta: \hat{H}^i(G, M \otimes N) \xrightarrow{\sim} \hat{H}^{i+1}(G, K \otimes N);$$

moreover, any cup product satisfying the analogue of V.3.3 is compatible with these isomorphisms, in the sense that

$$\begin{array}{ccc} \hat{H}^p(G, M) & \approx & \hat{H}^{p+1}(G, K) \\ \downarrow - \cup v & & \downarrow - \cup v \\ \hat{H}^{p+q}(G, M \otimes N) & \approx & \hat{H}^{p+q+1}(G, K \otimes N) \end{array}$$

commutes for any $v \in \hat{H}^q(G, N)$. Now suppose we have two cup products $\hat{H}^p \otimes \hat{H}^q \rightarrow \hat{H}^{p+q}$ as in the statement of the lemma. By hypothesis they agree when $p = q = 0$, so the square above allows us to prove by descending induction on p that they agree for $p \leq 0$ and $q = 0$. Similarly, writing N as a quotient of an induced module, we can extend this by descending induction on q to the case $p \leq 0, q \leq 0$. Next we embed M in an induced module and prove inductively that the cup products agree for p arbitrary and $q \leq 0$, and finally we embed N in an induced module to prove that they agree for all p and q . $\square$

Still assuming the existence of Δ , we can now easily prove the remaining properties of the cup product: The fact that $1 \in \mathbb{Z}/|G| \cdot \mathbb{Z} = \hat{H}^0(G, \mathbb{Z})$ is an identity is proved as in V.3.4 or by dimension-shifting [it is obviously true in dimension 0]; associativity is proved by dimension-shifting [it is obviously true in dimension 0]; commutativity can be deduced from 5.8 [$u \cup v$ and $(-1)^{pq} t_*(v \cup u)$ are two cup products with the required properties] or, alternatively, there is a direct proof analogous to that of V.3.6; finally, the fact that the cup product behaves properly with respect to restriction and co-restriction follows directly from the definition as in V.3.7 and V.3.8, or by dimension-shifting.

To complete the construction of the cup product, we must prove the existence of a complete diagonal approximation Δ . We will use the following three lemmas:

(5.9) Lemma. $F \hat{\otimes} F$ is acyclic and dimension-wise relatively injective.

PROOF. Relative injectivity follows from the fact that an arbitrary direct product of relative injectives is relatively injective. To prove acyclicity, let $h: F \rightarrow F$ be a contracting homotopy for F , regarded as a complex of abelian groups (cf. 3.2), and let $H = h \hat{\otimes} \text{id}_F$. I claim that H is a contracting homotopy for $F \hat{\otimes} F$ (regarded as a complex of abelian groups). Indeed, recalling the definition of the differential ∂ in $F \hat{\otimes} F$, we have

$$\begin{aligned} \partial H + H\partial &= (d \hat{\otimes} \text{id}_F + \text{id}_F \hat{\otimes} d) \circ (h \hat{\otimes} \text{id}_F) \\ &\quad + (h \hat{\otimes} \text{id}_F) \circ (d \hat{\otimes} \text{id}_F + \text{id}_F \hat{\otimes} d) \\ &= dh \hat{\otimes} \text{id}_F - h \hat{\otimes} d + hd \hat{\otimes} \text{id}_F + h \hat{\otimes} d \quad (\text{cf. exercise 8 of §I.0}) \\ &= (dh + hd) \hat{\otimes} \text{id}_F \\ &= \text{id}_F \hat{\otimes} \text{id}_F \\ &= \text{id}_{F \hat{\otimes} F}, \end{aligned}$$

whence the claim. $\square$

(5.10) Lemma. Let (C, ∂) and (C', ∂') be two acyclic chain complexes of $\mathbb{Z}G$ -modules. Assume that each C_i is projective and each C'_i is relatively injective. If